

**CCA EduSys: AI-Enhanced School Management System
with AI Assistance for Calvary Christian Academy**

A Thesis

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**CORONEL, KENT KENJI
BANGSOY, JOHN LUIZ
TOBIAS, CEDRICK JAY
TUMALOM, JOHN LOYD
TURREDA, KOBE BRYAN**

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KENT KENJI B. CORONEL
JOHN LUIZ V. BANGSOY
CEDRICK JAY A. TOBIAS
JOHN LOYD P. TUMALOM
KOBE BRYAN D. TURREDA
THE PROPONENTS

DEDICATION

This thesis study is wholeheartedly dedicated to those who have been a constant source of inspiration, strength, and support throughout our academic journey.

First and foremost, we dedicate this work to our beloved parents and families, whose unconditional love, encouragement, and sacrifices have made it possible for us to pursue and complete our education. Their guidance and support have been our foundation in overcoming every challenge along the way.

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To ourselves, we dedicate this work as a testament to our perseverance, hard work, and commitment in overcoming challenges and achieving this milestone in our academic journey.

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ABSTRACT

As Calvary Christian Academy experiences growth in its student population and administrative operations, the existing fragmented and manual system for managing student records, enrollment, tuition payments, and report generation has become inefficient, resulting in data redundancy, delays in processing, increased workload, and a higher risk of errors. This study developed an AI-Enhanced School Management System to streamline and improve the school's data management processes.

The system features a centralized, web-based platform that securely stores and manages student information, enabling faster access and improved accuracy. It incorporates AI Auto-Fill Enrollment to reduce repetitive data entry, AI Error Detection to identify inconsistencies, and an AI Report Generator to produce academic and financial reports instantly. A digital tuition tracking module was also implemented to enhance payment monitoring and record accuracy.

The implementation of the system improved operational efficiency, reduced administrative workload, minimized data redundancy, and enhanced the accessibility and reliability of records. It also enabled faster report generation and better monitoring of student and financial data, supporting more effective decision-making and overall school management.

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CHAPTER 1: INTRODUCTION

Artificial Intelligence (AI) has provided new opportunities to enhance information systems by assisting users in completing tasks more efficiently. AI-powered assistance can support information retrieval, provide intelligent responses, reduce repetitive work, and improve decision-making while allowing users to maintain control over administrative processes.

The continuous advancement of technology has influenced the education sector by improving how institutions manage academic, administrative, and financial processes. School management systems have become essential tools for organizing student information, streamlining workflows, and improving access to accurate data. However, many educational institutions still encounter challenges caused by manual processes, disconnected platforms, and increasing administrative responsibilities.

This study introduces CCA EduSys: AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy, a centralized platform designed to improve the management of student records, enrollment, tuition monitoring, academic information, and administrative processes. Through AI assistance integrated into different system modules, the proposed system aims to improve efficiency, accuracy, accessibility, and user experience among school administrators, staff, students, and parents.

Background of the Study

Calvary Christian Academy (CCA), located at Blk 1 Lot 9 & 10, Sarmiento Townville Subdivision, Barangay Poblacion I, City of San Jose Del Monte, Bulacan 3023, is a private educational institution established in 1992. The school offers basic education from Kindergarten to Grade 10, with approximately 30–32 students per section and one section for each grade level, resulting in an estimated total student population of 270–300 students. The school manages various operations involving student enrollment, academic records, tuition payments, and

communication with parents. Its workforce consists of approximately 20 teachers and administrative personnel, including the Principal, Registrar Staff, Cashier, IT Staff, Librarian, Guidance Counselor, and School Nurse, who handle different academic and administrative responsibilities.

CCA is committed to providing quality education that develops Christ-like leaders who are locally committed and globally competent. Its vision focuses on creating a learning environment that supports the academic and personal development of students, while its mission emphasizes strong partnership with parents, quality academic instruction, Bible-based teaching, critical thinking, extracurricular activities, and high standards in performance and behavior. The school also aims to develop students who demonstrate responsibility, integrity, independent thinking, and skills that prepare them for higher education, careers, and active participation in their community. These goals highlight the importance of providing an organized and effective school environment that supports students, parents, teachers, and administrative staff.

Currently, CCA manages student information using multiple platforms and manual procedures, resulting in repeated data entry, fragmented records, and difficulties in accessing information efficiently. Administrative staff spend approximately 10–15 hours per week managing and updating paper-based student records. The use of multiple storage locations makes information retrieval less efficient and requires additional time and effort when locating, checking, and updating records. Meanwhile, tuition payments are monitored using spreadsheets and physical receipts, which increases verification workload and contributes to payment record errors caused by manual tracking and updating. School-wide report preparation also requires approximately 10–12 hours, which may delay the availability of information needed for administrative decision-making and other school activities. These processes increase the workload of administrative staff and make it more difficult to maintain organized, accurate, and accessible information.

These current practices show the need for a more organized approach to managing school information while supporting CCA's commitment to quality education and strong partnerships with parents. A centralized system can help make important student and administrative information easier to access, maintain, and use. It can also support the school's goal of improving its operations while providing a more modern and accessible environment for staff, students, and parents.

To address these challenges, this study proposes CCA EduSys: AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy. The proposed system will provide a centralized platform that integrates student records, enrollment management, tuition monitoring, academic information, and reporting processes. AI assistance will be incorporated across different modules, including Registrar, Admission, Cashier, Student/Parent Portal, Principal Dashboard, and IT Administration, to support users through intelligent information retrieval, system guidance, and faster access to relevant data. The system aims to reduce redundant tasks, improve student record management, support faster report generation, and improve tuition monitoring while providing a more accessible and user-centered school management solution that aligns with the school's mission and goals.

Review of Related Literature, Studies, and Systems

a. Review of Related Literature

Foreign Literature

The integration of Artificial Intelligence (AI) and modern digital frameworks into educational management has become a focal point for institutional transformation, particularly in addressing the inefficiencies of traditional, fragmented systems. Recent advancements demonstrate that AI-driven Student Management Information Systems (AISMIS) can drastically optimize campus administration by shifting from decentralized, paper-based workflows to centralized, high-speed online platforms. These systems utilize big data analysis algorithms, such as

Random Forest, to process and categorize complex datasets with significant precision and speed (Wang, 2024).

The construction of an AI Student Management Information System using big data analysis algorithms demonstrates how advanced computational techniques can improve data processing and decision-making within educational environments. This system leverages big data analytics to efficiently manage large volumes of student information, enhancing accuracy and system responsiveness compared to traditional methods. Findings from the study suggest that integrating AI and big data capabilities into Management Information System (MIS) facilitates more effective handling of complex datasets, supports predictive insights, and strengthens institutional data management practices (Zhang et al., 2024).

Research on the application of student information management systems in higher vocational colleges reveals their important role in improving educational administration and operational efficiency. These systems facilitate the organization of student records, support academic tracking, and enable more effective coordination of institutional activities. The study found that when properly implemented, such systems enhance administrative workflows, improve data accuracy, and contribute to better institutional management practices (Huang, 2023).

The design and simulation of a computerized student information tracking system demonstrate the advantages of automating student record management through digital platforms. The study revealed that computerized systems improve the organization, retrieval, and monitoring of student information compared to traditional manual processes and that simulation tools can help validate system functionality before full implementation. Such systems contribute to improved data accuracy, faster processing, and support for administrative decision-making in educational institutions (Villanueva & Santos, 2025).

The implementation of AI combined with cloud computing for student management information security strengthens the protection of sensitive educational data against unauthorized access and cyber threats. AI-driven security mechanisms such as anomaly detection and real-time threat response enhance system robustness, while cloud computing enables scalable, centralized data storage. Research indicates that this combined approach improves system reliability, data integrity, and administrative control, making it an effective solution for securing student information systems (Zhang & Liu, 2023).

Local Literature

The transition from manual record-keeping to digitized, intelligent systems is a critical response to documented operational bottlenecks in Philippine educational institutions, where educators face significant challenges in manual data entry and information storage. These manual procedures often result in data inconsistencies, security risks, and slow retrieval speeds, necessitating the development of centralized platforms that can automate core functions like admissions and report generation to improve administrative efficiency (Falolo et al., 2022).

School Management Information Systems (SMIS) play a vital role in improving administrative efficiency through features such as data capturing, performance management, inquiry processing, and user management. These systems enable institutions to organize and analyze student and institutional data effectively, leading to better decision-making and streamlined operations. By centralizing information and automating processes, SMIS enhances accuracy, accessibility, and overall management performance in educational settings (Vargas, 2024).

The development of a Student Records Management System (SRMS) in Magalalag National High School highlights the need to replace manual recordkeeping with more efficient digital solutions. The study found that traditional paperbased systems lead to issues such as limited storage, difficulty in organizing documents, and increased workload. By implementing a computerized system

developed using the Software Development Life Cycle (SDLC), the institution improved data storage, retrieval, and overall system efficiency, while achieving a high level of software quality standards (Baliwagan et al., 2024).

The development of IntelliSchool, a student information system designed for senior high schools, demonstrates the effectiveness of digital solutions in managing academic data and improving school operations. The system provides a centralized and paperless platform that ensures data accuracy, automates grade computation, and enhances communication between teachers, students, and parents. Findings showed high usability and efficiency, indicating that such systems can significantly improve data management and overall institutional performance (Campanan et al., 2024).

The Student Information System developed for senior high school at Bestlink College of the Philippines demonstrates the effectiveness of digital platforms in managing student data and improving academic processes. The system allows users to access real-time information such as grades, attendance, and schedules, making data management more efficient and accessible for both students and faculty. Additionally, the implementation of the system reduced manual processing and improved the overall organization and monitoring of student records, contributing to better administrative performance (Barbero et al., 2024).

b. Review of Related Studies

Foreign Studies

Recent developments in educational technology have introduced AI-driven school management systems aimed at improving administrative efficiency and student performance monitoring. A study found that a React-based system integrated with AI algorithms effectively analyzed student data, reduced administrative workload, and improved user experience. It also enhanced the accuracy of performance analytics, supporting better decision-making and personalized learning. These

findings highlight the advantages of AI-based systems over traditional school management software (Dwivedi et al., 2023).

Education Management Information Systems (EMIS) serve as centralized platforms for storing and analyzing student data, enabling educators and administrators to monitor academic performance and support data-driven decision-making. These systems provide real-time access to information, allowing timely interventions and personalized learning experiences. They also improve instructional planning, resource allocation, and administrative efficiency through features such as enrollment tracking, attendance monitoring, and analytics dashboards. Overall, EMIS plays a key role in modernizing education by integrating with other digital and AI-driven tools to enhance teaching and learning outcomes (Organization for Economic Co-operation and Development (OECD), 2023).

Recent studies have emphasized the use of artificial intelligence (AI) in student management systems for monitoring academic performance and predicting at-risk students. A hybrid AI model combining decision trees, random forests, support vector machines, and neural networks showed improved adaptability and predictive accuracy across multiple datasets. The results demonstrated higher performance compared to individual models, achieving up to 98.8% accuracy and reducing error rates, making it effective for early intervention and academic monitoring (Wang, 2025).

A systematic review examined how artificial intelligence (AI) enhances student engagement in higher education through effective teaching methods. An analysis of 73 studies found that AI tools, such as chatbots and adaptive systems, are most effective when combined with interactive pedagogies like flipped classrooms and project-based learning. The study introduced a model highlighting how AI and teaching strategies influence student engagement, while also noting concerns

related to ethics, privacy, and equity. These findings stress the importance of proper instructional integration in maximizing AI's impact (Long et al., 2025).

The integration of artificial intelligence (AI) in education has been found to improve student personnel management and administrative decision-making. A study in secondary schools showed that AI enhances data handling, resource allocation, and overall administrative efficiency. The findings indicate that AI supports school leaders in making more informed and responsive decisions, contributing to improved educational management. The study recommends promoting AI adoption through training and infrastructure support (Barbero et al., 2024).

Local Studies

A qualitative study explored the experiences of school administrators integrating AI into educational management. Findings indicated that AI streamlines routine tasks, allowing leaders to focus on instruction and stakeholder engagement, while providing real-time insights into student performance, enrollment, and operations. Successful adoption requires adequate infrastructure, staff training, and attention to transparency, bias, and privacy, demonstrating AI's potential to support evidence-based and equitable decision-making (Mapula, 2025).

This study examined the impact of advanced technologies, including big data analytics, cloud computing, and communication platforms, on school management. Findings showed that these technologies significantly improved management efficiency and supported effective decision-making. The study emphasizes the need for professional development and strategic digital resource allocation to ensure successful technology integration, providing guidance for policymakers and educators seeking to enhance school management practices (Ramos & Bauyot, 2025).

A School Management Information System (SMIS) was developed for a community college to streamline administrative tasks such as admissions, data processing, and report generation. Using Agile methodology and evaluated with

the ISO 25010 quality model, the system demonstrated high functionality, usability, and reliability, effectively enhancing administrative efficiency. The study recommended future improvements, including mobile attendance monitoring and on-campus kiosks, to further optimize service delivery (Abasolo & Reyes, 2021).

A study examined the perceptions and usage of AI tools, such as Chat GPT and Grammarly, among DepEd personnel in Tandag City. Findings showed high familiarity and positive attitudes toward AI, with Chat GPT being the most used tool, although formal training was limited. Participants acknowledged AI's potential to enhance creativity and productivity while noting concerns about data privacy, job security, and reliability. The study recommends professional development on AI applications, ethical use, and data security, as well as further research on AI's longterm educational impact (Sibug, 2026).

A study at the Asian Institute of Computer Studies Senior High School examined the impact of custom digital tools on student enrollment efficiency. Findings showed that an online enrollment system streamlined data entry, verification, and processing, reducing administrative workload and improving transaction efficiency compared to traditional manual methods (Ramos, Conejos, Rey, Dalman, & Gundran, 2021).

c. Review of Related Systems



Figure 1. *VaultEdu Academy User Interface.*

The VaultEdu PH: AI-Powered School Management System is a cloud-based educational management platform developed to streamline academic and administrative operations within educational institutions. The system centralizes essential school functions, including student enrollment, attendance monitoring, grading, scheduling, financial management, and communication. It also integrates artificial intelligence (AI) technologies that automate repetitive administrative tasks, generate predictive analytics, and provide interactive dashboards to support data-driven decision-making. Furthermore, the platform improves operational efficiency by reducing manual processes and enhancing collaboration among school administrators, teachers, students, and parents. However, institutions may need additional customization depending on their specific operational requirements (VaultEdu PH, n.d.).

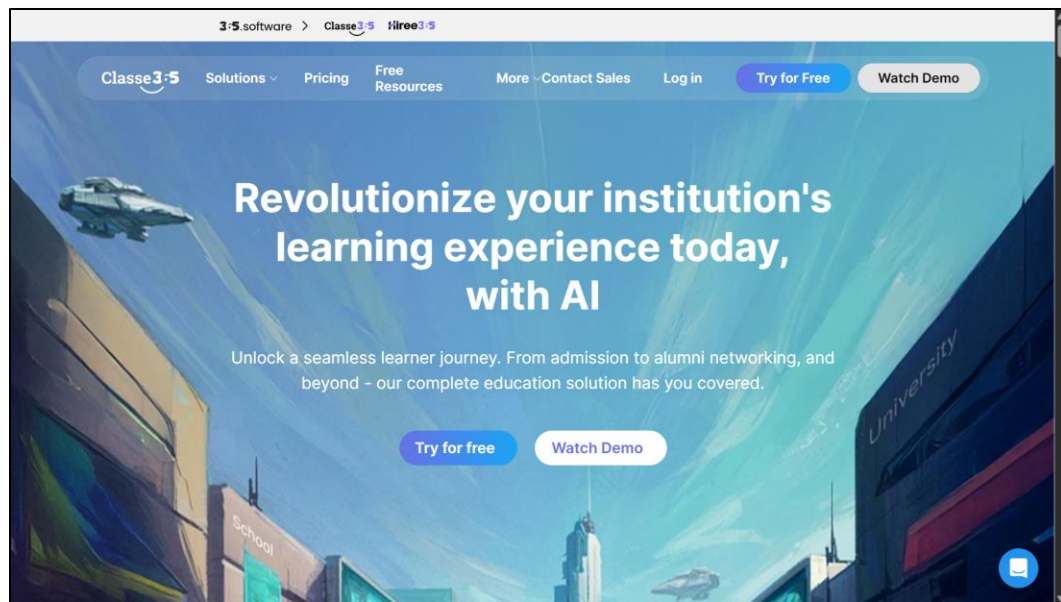


Figure 2. *Classe365 User Interface.*

Classe365: Unified Student Information Management System is a centralized digital *database* that streamlines admissions, attendance, and grading while facilitating seamless communication between schools and parents. The platform automates repetitive administrative tasks and integrates financial tools for fee collection and academic performance analytics to support data-driven decision-making. (Classe365, 2025).

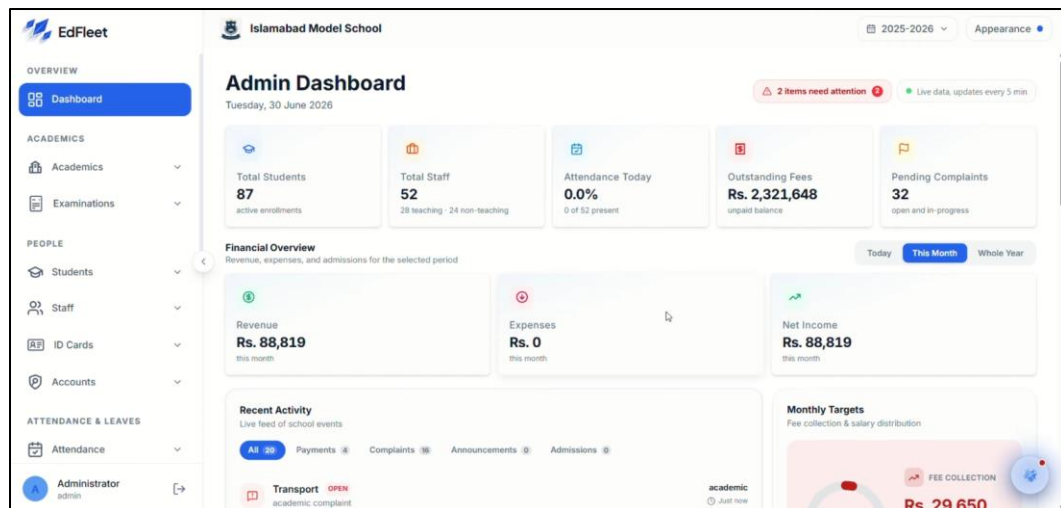


Figure 3. EdFleet User Interface.

The EdFleet: AI-Powered School Management Software is an integrated educational management platform designed to simplify school administration through automation and artificial intelligence. The system provides modules for student information management, attendance tracking, examination management, fee collection, academic reporting, and communication. It incorporates AI-powered features such as an intelligent virtual assistant, automated lesson planning, and smart reporting tools that reduce administrative workload and improve collaboration among teachers, students, parents, and school administrators. The platform enhances operational efficiency by digitizing routine school processes and supporting effective decision-making within educational institutions (Kabeer, 2026).

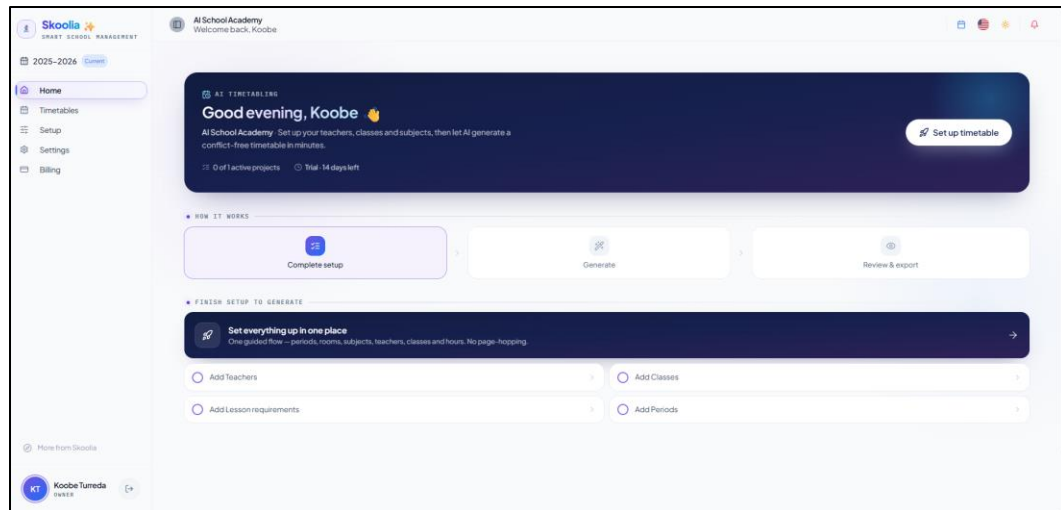


Figure 4. Skoolia User Interface.

The Skoolia: AI-Powered School Management System is a modern educational management platform that integrates artificial intelligence to improve academic planning and school administration. The system provides comprehensive modules for admissions, attendance monitoring, grading, examination management, timetable scheduling, financial management, and communication. It features AI-powered capabilities such as an intelligent timetable generator, automated report card comments, and AI-assisted academic support, helping schools optimize scheduling, reduce administrative workload, and improve overall operational efficiency. The platform offers a centralized solution for managing educational activities while supporting data-driven decision-making and enhanced collaboration among school stakeholders (Amer, 2025).

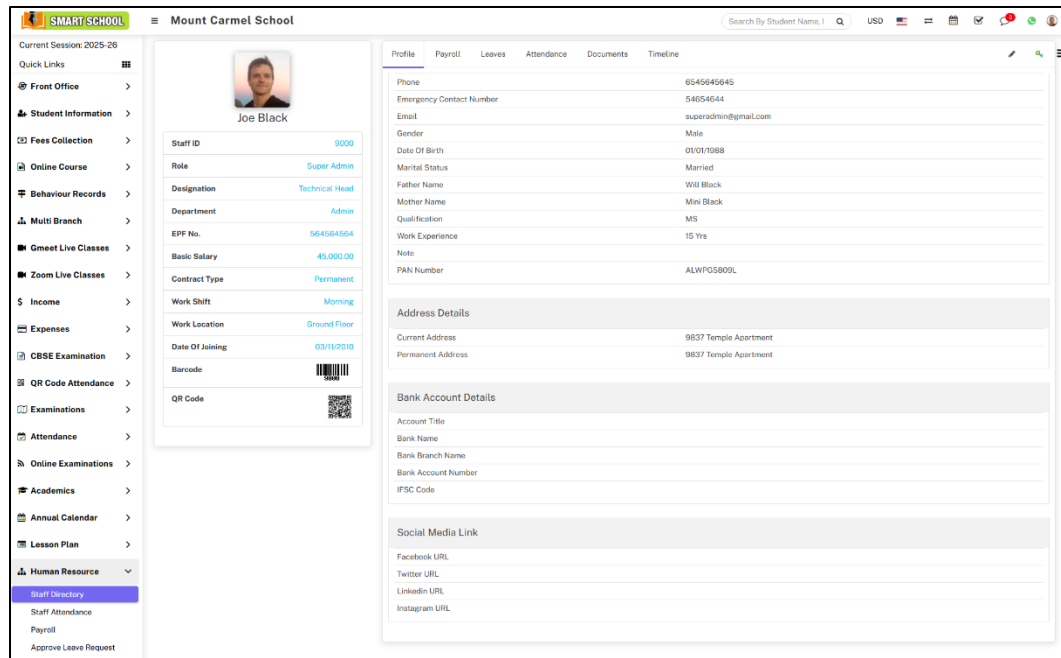


Figure 5. Smart School User Interface.

The Smart School: School Management System developed by QDOCS is an educational management platform designed to automate and centralize various academic and administrative processes. The system features an integrated interface that allows administrators, teachers, students, and parents to manage school-related activities efficiently. It includes modules for student information management, attendance monitoring, fee collection, examination management, timetable scheduling, library management, and report generation. Moreover, the system supports multiple user roles, enabling different stakeholders to access functions based on their responsibilities. The implementation provides a comprehensive solution for reducing manual administrative tasks and improving the organization of school operations; however, additional customization may be required to meet the specific requirements of individual educational institutions (QDOCS, n.d.).

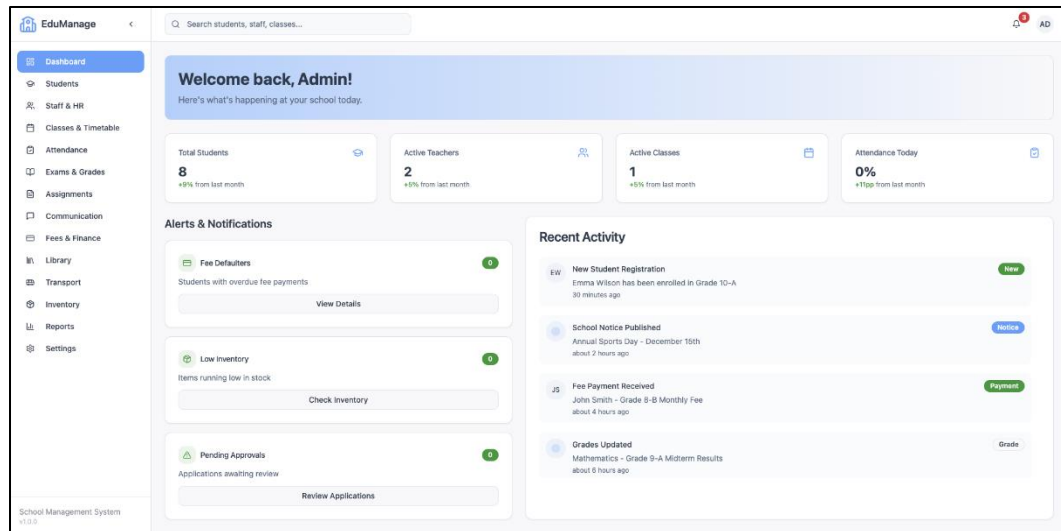


Figure 6. EduManage User Interface.

EduManage: School ERP System developed by Samyotech Software Solutions Pvt Ltd is an integrated educational management platform designed to automate and simplify academic and administrative operations. The system provides a centralized interface that enables schools to manage student records, staff information, attendance, examinations, assignments, fees, communication, library services, transportation, inventory, and reporting activities. Furthermore, the system supports digital learning and school administration by integrating multiple modules into a single platform, allowing educational institutions to improve operational efficiency and data organization. The implementation utilizes modern web technologies and provides a comprehensive solution for managing various school processes; however, further customization may be required to satisfy the specific requirements and workflows of different educational institutions (Samyotech Software Solutions Pvt Ltd, n.d.).

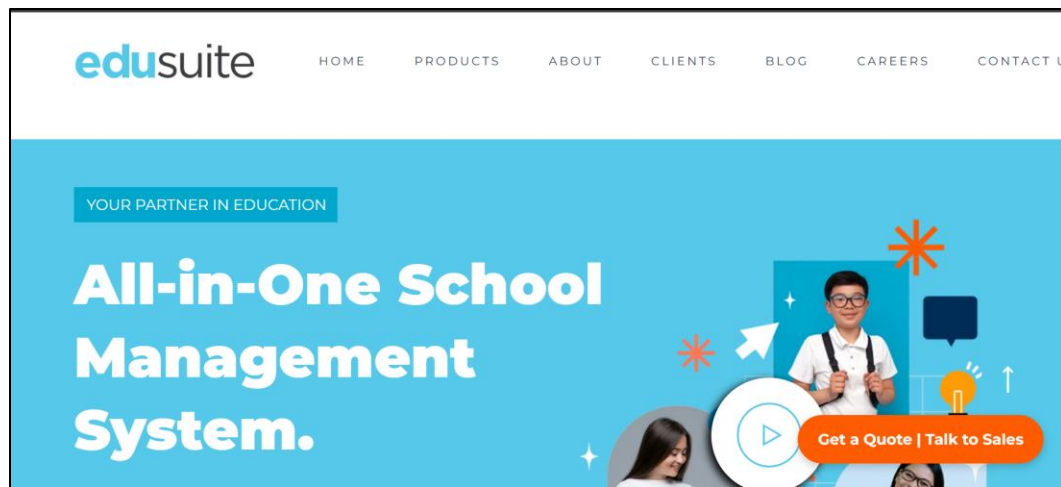


Figure 7. Edusuite User Interface.

Edusuite: AI-Driven School Management Solution is a cloud-based system that utilizes smart *algorithms* and AI to analyze campus data for optimized planning and operations. Its core features include automated student advising, predictive forecasting of class demand, and a built-in automated scheduling maker that reduces processing time for registration and student payments. (Edusuite, 2025).

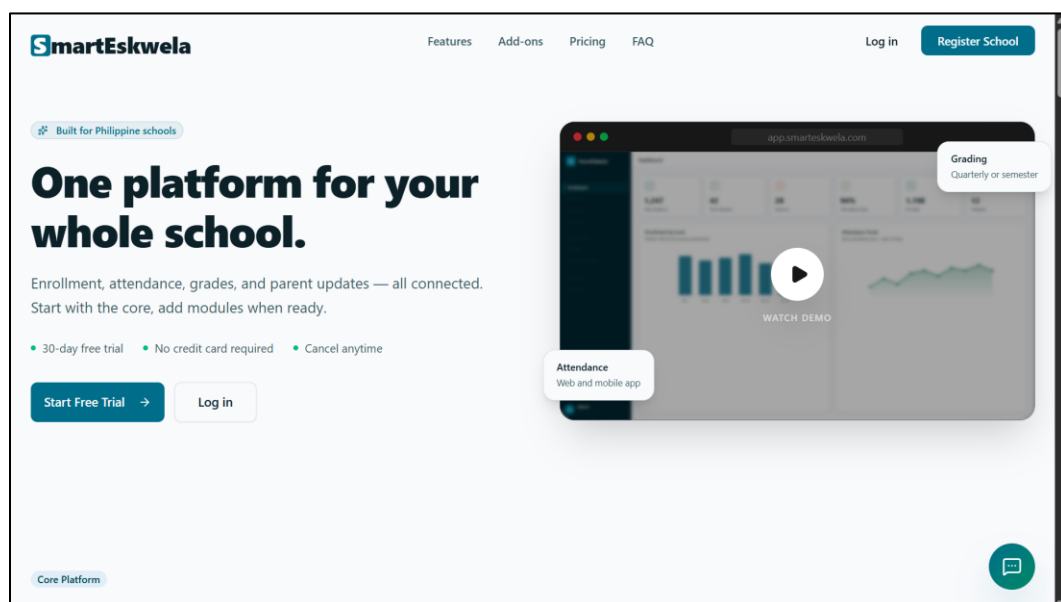


Figure 8. Smart Eskwela User Interface.

Smart Eskwela: School Management Platform is an integrated platform designed for Philippine schools that connects enrollment, attendance, and grading into a single web and mobile interface. It offers modular add-ons such as *biometric* attendance and online fee collection, providing school heads with clear visibility into attendance trends and enrollment counts via interactive dashboards. (Smart Eskwela, 2026).

Synthesis

The reviewed school management systems, including VaultEdu PH, Classe365, EdFleet, Skoolia, Smart School, EduManage, Edusuite, and Smart Eskwela, demonstrate how digital platforms improve educational administration through centralized data management, automation, and improved accessibility of school information. These systems provide essential features such as student information management, enrollment processing, attendance monitoring, financial management, reporting, and communication tools. Furthermore, the integration of artificial intelligence features such as virtual assistants, automated reporting, predictive analytics, and intelligent data processing shows the potential of AI in reducing administrative workload and supporting faster decision-making within educational institutions.

However, most existing systems are designed as general solutions and may require further customization to meet the specific operational needs of individual schools. While some platforms provide AI-powered features, these are often limited to specific functions rather than providing continuous AI assistance across multiple user roles and school processes. To address these limitations, this study proposes CCA EduSys: AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy, which focuses on integrating centralized school management with AI assistance across different modules, including Registrar, Admission, Cashier, Student/Parent Portal, Principal Dashboard, and IT Administration. The proposed system aims to improve

information accessibility, reduce repetitive tasks, support users in completing their responsibilities, and enhance the overall efficiency of school operations.

Statement of the Problem

Calvary Christian Academy uses different systems to manage student records, academic information, and tuition payments. Since information is stored across several platforms and records, staff may need to enter the same data more than once and use different sources when looking for important information. This can make it harder to keep records accurate, organized, and easy to access. Tuition payments are monitored through spreadsheets and physical receipts, which makes checking and updating payment records more time-consuming. The use of different storage locations also makes it difficult to retrieve information and prepare reports efficiently. These practices increase the workload of staff and may delay the processing of records, reports, and other administrative tasks.

a. General Problem

Calvary Christian Academy faces difficulties in managing student records, tuition payments, and report generation efficiently. The use of multiple systems and manual processes results in repeated data entry, increased administrative workload, and delays in accessing, updating, and preparing important information. These difficulties make it harder for staff to keep records organized, accurate, and easy to access.

b. Specific Problems

- Student information is recorded repeatedly in approximately 6–10 different forms and documents per student, depending on the number of documents required for each student. Administrative staff spend around 10–15 hours per week managing paper-based student records. This results in repeated data entry, increased workload, and a higher risk of errors or inconsistencies in student information.

- Accessing student records takes approximately 10–15 minutes per request because student records are stored in physical master files and spreadsheets. This makes locating, checking, and retrieving the needed information difficult and time-consuming.
- The preparation of school-wide academic and financial reports requires approximately 10–12 hours, leading to delays in releasing information and supporting administrative decision-making.
- Tuition payments are currently tracked using spreadsheets and physical receipts, increasing administrative workload and the likelihood of verification errors, with an estimated five (5) payment-record errors occurring per month.

Objectives of the Study

a. General Objective

To develop a School Management System with AI-assisted features that improves the management of student records and report generation at Calvary Christian Academy by reducing redundant data entry, administrative workload, and delays in accessing, updating, and preparing important information, while also improving tuition payment monitoring.

b. Specific Objectives

- To reduce data redundancy and administrative workload by implementing a centralized student information database that stores student records once, minimizes repeated data entry, and reduces paper-based record management time from 10–15 hours per week to 2–5 hours per week while improving the accuracy and accessibility of student information across all system modules.
- To improve the accessibility of student records and tuition information by implementing a centralized system that reduces retrieval time from 10–15 minutes to 1–5 minutes per request and integrating an AI-powered

assistant to help users quickly search for records, answer record-related inquiries, and facilitate faster access to information.

- To automate the generation of academic and financial reports, reducing report preparation time from 10–12 hours to 5–10 minutes per report, and integrate an AI-powered assistant that assists users in retrieving, summarizing, and interpreting report information for faster and more informed decision-making.
- To enhance the accuracy and efficiency of tuition payment monitoring by implementing a digital payment tracking system integrated with centralized student records, reducing payment-record errors from five (5) per month to 0–1 per month, and integrating an AI-powered assistant to help users retrieve payment

Hypothesis / Assumption

The study proposes that the implementation of CCA EduSys: AI-Enhanced School Management System with AI Assistance can significantly improve the efficiency and accuracy of school administrative processes at Calvary Christian Academy. By integrating a centralized student information database, digital tuition payment monitoring, automated report generation, and AI-assisted information retrieval, the proposed system is expected to reduce redundant data entry, administrative workload, record retrieval time, report preparation time, and tuition payment-record errors. The study expects that the implementation of the system will contribute to more organized, accessible, accurate, and efficient management of student and administrative information.

Theoretical Framework

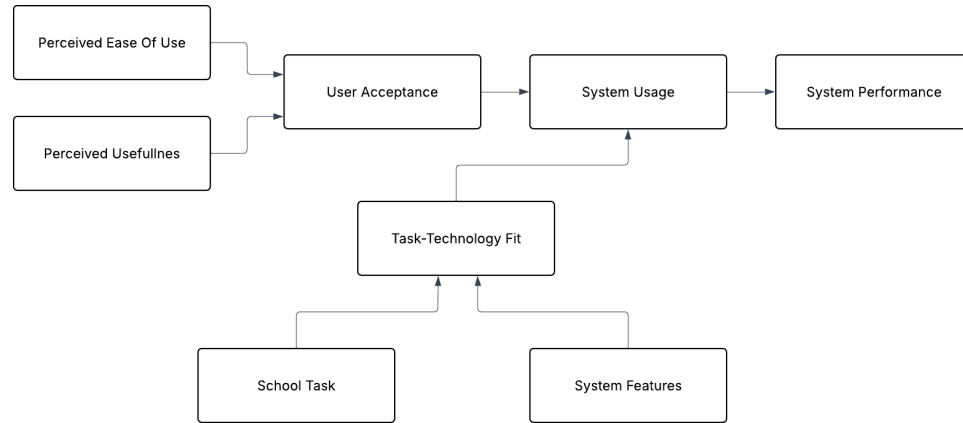


Figure 9. Technology Acceptance Model (TAM) & Task-Technology Fit (TTF) Theory.

This study is anchored on the Technology Acceptance Model (TAM) and Task-Technology Fit (TTF) Theory. These theories help explain how CCA EduSys: AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy can improve school operations and support users in adopting and using the system.

The Technology Acceptance Model (TAM) by Fred D. Davis explains that users are more likely to accept and use a system when they find it easy to use and useful. In this study, **Perceived Ease of Use** refers to how easily school personnel, such as the Principal, Registrar Staff, Cashier, Admission Staff, and other authorized users, can use CCA EduSys to manage enrollment, student records, tuition payments, and reports. **Perceived Usefulness** refers to how the system helps users complete their work more efficiently by reducing repeated data entry, improving access to student information, supporting tuition monitoring, and making report preparation faster. The AI assistance of the system can also help users find information, receive system guidance, and complete tasks more efficiently.

The Task-Technology Fit (TTF) Theory by Goodhue and Thompson states that technology can improve work performance when its features match the tasks that users need to perform. In this study, school personnel perform tasks such as enrollment processing, student record management, tuition monitoring, payment verification, and report preparation. CCA EduSys is designed to support these tasks through a centralized student information system, integrated modules, and AI-assisted features. These features can help reduce repeated data entry, improve record accuracy, make information easier to retrieve, and reduce the time needed to complete administrative tasks.

By combining TAM and TTF, the study explains user acceptance, ease of use, usefulness, and task support. Together, these theories guide the development of a user-friendly CCA EduSys.

Conceptual Framework



Figure 10. Input Process Output (IPO) Chart.

The conceptual framework of this study follows the **Input-Process-Output (IPO) model**, with a feedback loop that supports continuous improvement of the system. It presents the development of **CCA EduSys: AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy** using the Agile Model. The framework is divided into three main parts: Input, Process, and Output, where each part contributes to the development of an effective and user-friendly school management system.

The **Input** phase includes the knowledge, software, and hardware requirements needed to develop and operate CCA EduSys. The knowledge requirements include web development technologies, database tools, backend technologies, and AI and machine learning tools. The software requirements include the operating system, development tools, programming environments, and web browsers, while the hardware requirements provide the necessary computer equipment and internet connection. These resources serve as the foundation for developing and implementing the proposed system.

The **Process** phase follows the six phases of the Agile Model: **Planning, Analysis, Design, Development, Testing, and Deployment**. The researchers first identify the school's needs and existing problems, analyze the required system features, design the system structure and interface, and develop the different modules with AI assistance. The system is then tested to ensure its functionality, accuracy, and usability before deployment. The **Output** is CCA EduSys, which provides centralized student records, improved enrollment and tuition management, AI-assisted school operations, and automated reporting. User feedback and system performance can also be used for future updates and improvements, helping ensure that the system continues to meet the needs of Calvary Christian Academy.

Scope and Delimitation of the Study

a. Scope of the study

- This study focuses on the design, development, and implementation of CCA EduSys: An AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy. The system provides a centralized, web-based platform for managing student information, enrollment, academic records, tuition payment records, report generation, and administrative processes.
- The proposed system includes six major modules: the Registrar Module, Admission Module, Cashier Module, Student/Parent Portal, Principal Dashboard, and IT Administration Module. These modules are designed to streamline school operations while providing role-based access to authorized users.
- The Teacher Module enables teachers to access their assigned class lists, manage student grades, monitor academic records, and submit academic information within the system. It also provides AI assistance for information retrieval and answering system-related inquiries to help teachers perform their administrative tasks more efficiently.
- The Registrar Module enables authorized personnel to manage student records, academic information, and record retrieval through a centralized database. It also provides AI-assisted search to improve the accessibility and retrieval of student information.
- The Admission Module supports online enrollment, student information collection, document management, and AI-assisted enrollment guidance. These features are intended to simplify enrollment procedures and reduce processing time.
- The Cashier Module manages tuition payment records, payment monitoring, and AI-assisted payment inquiries. This module improves payment tracking and facilitates faster verification of student payment information.

- The Student/Parent Portal allows students and parents to access student information, academic records, and tuition payment status. It also provides AI assistance to answer common inquiries and improve access to school-related information.
- The Principal Dashboard provides access to academic, enrollment, and monitoring school operations. AI-assisted report summarization supports administrators in reviewing information more efficiently for decision-making.
- The IT Administration Module is responsible for user account management, system maintenance, security, and access control. Only authorized IT personnel can create, modify, suspend, reactivate, reset passwords, and assign user roles for system users.
- Artificial Intelligence is integrated into the system as an assistive technology that supports intelligent search, natural language inquiries, information retrieval, process guidance, report summarization, and user assistance. The AI features are intended to improve user productivity and operational efficiency without replacing human decision-making.
- The system will be implemented and evaluated within Calvary Christian Academy to assess its functionality, usability, reliability, and effectiveness. The evaluation will be guided by the Technology Acceptance Model (TAM) and Task-Technology Fit (TTF) framework to measure user acceptance and task support.

b. Delimitation of the Study

- The study is limited to the development of a web-based school management system for Calvary Christian Academy. It focuses on student information, enrollment, academic records, tuition payment records, report generation, and user account management.
- The AI assistance is limited to intelligent search, natural language inquiries, information retrieval, enrollment guidance, report

summarization, and general user assistance. It does not include predictive analytics, automated grading, behavioral analysis, recommendation systems, or autonomous decision-making.

- The system does not connect to external payment gateways, learning management systems, government education databases, or other third-party platforms. Tuition monitoring is limited to recording and managing payment information within CCA EduSys.
- The system is designed specifically for Calvary Christian Academy and requires a stable internet connection.

Significance of the Study

a. Contribution to Knowledge

This study is important for several groups, including students, cashiers, registrars, teachers, parents, school staff, school administrators, technology developers, and future researchers.

By improving the current student information and academic monitoring process, the research aims to make record management faster, more accurate, and more reliable.

b. Beneficiaries of the Study

Calvary Christian Academy: Calvary Christian Academy will be the primary beneficiary of this study through the implementation of an AI-Enhanced School Management System. The system will streamline school operations by centralizing data, reducing redundancy, and improving the accuracy and efficiency of record management, enrollment, tuition tracking, and report generation.

Departments of Calvary Christian Academy (Cashier & Registrar): The Cashier and Registrar departments will benefit from automated and organized processes in handling student information and financial records. The system

will minimize repetitive data entry, enhance accuracy in tuition monitoring, and accelerate enrollment procedures, thereby improving overall efficiency and reducing administrative workload.

Faculty and Administrative Staff of Calvary Christian Academy: Faculty members and administrative staff will benefit from improved accessibility to student data and automated reporting tools. The system will enable faster retrieval of academic and financial records, allowing personnel to focus more on instruction, student monitoring, and timely academic interventions.

Students of Calvary Christian Academy: Students will benefit indirectly from the improved efficiency and accuracy of school services. The system will support timely monitoring of academic performance through AI-assisted tools, helping identify students who may require additional academic support and contributing to enhanced learning outcomes.

Parents of Calvary Christian Academy: Parents will benefit from improved communication and easier access to their children's academic, attendance, and financial information. Through the system, parents can receive timely updates regarding enrollment status, attendance records, tuition balances, announcements, and school activities. This accessibility may strengthen parental involvement in their children's education and promote better collaboration between the school and families.

Other Educational Institutions: Other educational institutions may utilize this study as a reference for developing or enhancing their own school management systems. The proposed solution demonstrates how AI technologies can be integrated to address common challenges in data management, reporting, and administrative efficiency.

Technology Developers: Technology developers will benefit from the study as it presents a practical application of Artificial Intelligence, including AI

Assistance. It may serve as a guide for designing innovative systems tailored to educational environments.

Students: Students in Computer Science and related courses may use this study as a reference for understanding system development processes, database management, and AI integration. It can support their academic learning and future project development.

Future Developers: Future developers can use this study as a foundation for enhancing AI-based school management systems. They may expand the system's features, improve scalability, and adapt it for broader or more complex institutional requirements.

Future Researchers: Future researchers and proponents may benefit from this study by using it as a basis for further research and innovation. They may explore additional functionalities, integrate more advanced AI techniques, or apply the system in different educational contexts.

Definition of Terms

Academic Records (n.) /ak-uh-dem-ik rek-erdz/ — Student grades and other academic information that are recorded, managed, and accessed by authorized users through CCA EduSys.

Admission Module (n.) /ad--un mod-yool/ — A system module that manages student admission and enrollment information, including the collection, submission, and review of required student documents.

Artificial Intelligence (AI) (n.) /ar-tuh-fish-ul in-tel-uh-jens/ — A technology integrated into CCA EduSys to assist users with information retrieval, inquiries, process guidance, and report summarization.

AI Assistance (n.) /ay-eye uh-sis-tuhns/ — An integrated system feature that provides users with AI-supported help in retrieving information, answering inquiries, and completing supported school-related tasks.

AI-Assisted Search (n.) /ay-eye uh-sis-tid serch/ — A feature that uses AI to help authorized users find relevant student and school information within the system.

CCA EduSys (n.) /see-see-ay ed-yoo-sis/ — The proposed AI-enhanced school management system developed for Calvary Christian Academy to manage student records, enrollment, academic information, tuition records, and administrative processes.

Cashier Module (n.) /ka-sheer mod-yool/ — A system module that manages student tuition payment records and supports the monitoring and verification of payment information.

Centralized Student Information Database (n.) /sen-truh-lyzd stoo-dent in-fer-may-shun day-tuh-bays/ — A shared database within CCA EduSys that stores student information in one location for consistent access and management by authorized users.

Enrollment Management (n.) /en-rohl-ment man-ij-ment/ — The system process for recording, submitting, reviewing, and managing student enrollment information and supporting documents.

Information Retrieval (n.) /in-fer-may-shun ri-tree-vuhl/ — The process of locating and accessing student, academic, enrollment, or financial information stored in CCA EduSys.

IT Administration Module (n.) /eye-tee ad-min-uh-stray-shun mod-yool/ — A system module used to manage user accounts, assigned roles, access permissions, password-related functions, and system security.

Principal Dashboard (n.) /prin-suh-puhl dash-bord/ — A system interface that allows the principal to monitor academic, enrollment, financial, and other relevant school information through centralized system data.

Registrar Module (n.) /rej-uh-strar mod-yool/ — A system module used by authorized registrar personnel to manage student records, enrollment information, academic information, and related documents.

Role-Based Access Control (n.) /rohl-bayst ak-ses kun-trohl/ — A security feature that assigns system access according to the user's role, allowing authorized users to access only the functions and information relevant to their responsibilities.

Student/Parent Portal (n.) /stoo-dent pair-ent por-tuhl/ — A system interface that allows authorized students and parents to access relevant student information, academic records, and tuition payment status.

Teacher Module (n.) /tee-cher mod-yool/ — A system module that allows teachers to access assigned class lists, manage grades, monitor academic records, and submit academic information.

Tuition Monitoring (n.) /too-ish-un mod-i-ter-ing/ — The system process of tracking and verifying student tuition payment information through centralized payment records.

Web-Based System (n.) /web-bayst sis-tum/ — CCA EduSys as a browser-accessed platform that allows authorized users to manage and retrieve school information through its centralized system.

CHAPTER 2: METHODOLOGY

System Analysis

This section focuses on analyzing the existing administrative and record-keeping processes of Calvary Christian Academy (CCA) and identifying the root causes of inefficiency in student record management, tuition tracking, and report generation. Based on the results of interviews and observations conducted among the principal, registrar, teachers, and administrative staff, several recurring issues were identified. These include redundant entry across multiple forms, slow access to student records and tuition information due to fragmented storage, difficulties in updating student information such as changes in address, time-consuming report generation, manual tuition tracking using spreadsheets and physical receipts, and delays in enrollment processing. These findings revealed the need for a centralized, AI-enhanced school management system capable of reducing repetitive entry, assisting in error detection, improving record accessibility, and automating report generation. Thus, the proposed CCA EduSys: AI-Enhanced School Management System was conceptualized to address these specific issues and modernize the overall administrative operations of Calvary Christian Academy.

a. Requirements Documentation

The requirements documentation outlines the essential features and system constraints necessary to address the problems identified in the current administrative processes of Calvary Christian Academy (CCA). The proposed CCA EduSys: AI-Enhanced School Management System is designed with two key categories of requirements: functional and non-functional. These requirements were identified based on the findings gathered through interviews with the Registrar Administrator, questionnaires, observations of existing procedures, and review of relevant school documents. The system is intended to centralize student information, reduce repetitive administrative tasks, improve record management, and support more efficient enrollment, tuition monitoring, and report preparation.

The functional requirements define the core operations and services that CCA EduSys must provide. The student enrollment and registration feature allows authorized personnel to record and manage student information while reducing repeated encoding of the same data. The student records management feature provides a centralized database for storing, updating, and retrieving student information, allowing authorized users to access organized records when needed. The document management feature supports the submission, checking, and verification of required enrollment documents, reducing the reliance on manual document handling. The tuition and payment monitoring feature allows authorized personnel to record and monitor student payments and maintain organized financial information, reducing dependence on spreadsheets and physical receipts. The reporting feature generates organized reports related to student records, enrollment, and tuition information, reducing the time required for manual report preparation. Administrative management tools also allow authorized users to manage system information and user access according to their assigned responsibilities. In addition, the AI assistance feature supports selected administrative tasks and information-processing activities to help reduce repetitive work and improve the efficiency of school personnel.

The non-functional requirements focus on the quality, security, and performance of the system. Usability ensures that the system provides a simple, clear, and user-friendly interface that can be easily used by authorized school personnel. Reliability ensures that student, enrollment, and payment information remains accessible and that the system performs consistently during regular administrative operations. Security protects sensitive student and financial information through user authentication, access controls, and appropriate protection of stored data. Performance requires the system to process requests and retrieve records efficiently to minimize delays during enrollment, record management, payment monitoring, and report generation. Maintainability ensures that the system can be updated, corrected, and improved when new requirements or administrative needs

arise. Scalability allows the system to accommodate an increasing number of students, records, users, and transactions as CCA grows.

Overall, the identified functional and non-functional requirements provide the foundation for the development of CCA EduSys. By centralizing student records, streamlining enrollment and document management, improving tuition and payment monitoring, supporting automated report generation, and incorporating AI assistance, the proposed system aims to reduce repetitive administrative work and improve the accuracy, organization, and efficiency of school information management. These requirements ensure that CCA EduSys is not only capable of addressing the current administrative challenges of Calvary Christian Academy but is also designed to provide a reliable, secure, usable, and adaptable system for future administrative needs.

b. Current System Utilized

At present, Calvary Christian Academy (CCA) utilizes a combination of paper-based forms, spreadsheets, separate digital files, and existing administrative procedures to manage school operations, particularly in the Registrar and Cashier offices. Student records, enrollment documents, tuition payments, and other administrative information are handled through these different methods. Staff members manually record, update, and retrieve information as needed. While these methods allow the school to carry out its regular administrative operations, the information is not fully centralized or integrated into a single system, making it more difficult to efficiently manage and maintain student and administrative records.

Although the current approach is familiar and accessible to staff, it has several limitations in managing school information. Updating student information, such as a change of address, can be difficult and time-consuming because records may need to be checked and updated across different files or documents. Tuition monitoring through spreadsheets and physical receipts also requires additional

effort when tracking payments and preparing reports. These limitations can lead to repeated data entry, duplicated information, difficulty in tracking updated records, and delays in processing administrative transactions. Consequently, the current system may become less efficient as the number of students, records, and administrative transactions increases, creating a need for a more centralized and organized approach to school information management.

The New System (System Design)

a. Requirement Analysis

Based on the existing administrative processes of Calvary Christian Academy (CCA), interviews, observations, questionnaires, and review of school documents, the requirements for the proposed CCA EduSys: AI-Enhanced School Management System with AI Assistance were identified. The system requires knowledge of web development technologies, database tools, backend technologies, and artificial intelligence and machine learning tools to support its development and operation. It is also designed to address the identified concerns in student records, enrollment, tuition management, and report generation by providing centralized student records, improved information management, AI-assisted school operations, and automated reporting. These requirements serve as the foundation for developing a system that is responsive to the administrative needs of CCA.

The functional requirements include centralized student records management, enrollment management, tuition and payment management, AI-assisted school operations, and automated report generation. The centralized records feature allows authorized personnel to store, access, and update student information more efficiently, including necessary changes such as a student's address. Enrollment management supports the organization of student enrollment information, while tuition management provides a more organized approach to monitoring payments. The AI assistance feature supports selected administrative tasks and information

processing, while automated reporting helps reduce the time and effort required to prepare reports.

Meanwhile, the non-functional requirements emphasize usability, reliability, security, performance, maintainability, and scalability. The system should provide a user-friendly interface that allows authorized personnel to perform administrative tasks efficiently. It should maintain reliable access to student and administrative records while protecting sensitive information through appropriate security measures. The system should also provide efficient processing and retrieval of information, be maintainable for future improvements, and be scalable to accommodate the growing number of students, records, and administrative transactions of CCA.

To support the development and operation of CCA EduSys, the required software includes an operating system, development tools, programming environments, and web browsers, while the hardware requirements include the necessary computer equipment and internet connection. These requirements will be applied throughout the Agile Model's six phases: Planning, Analysis, Design, Development, Testing, and Deployment. Through these requirements and development processes, CCA EduSys aims to provide a centralized, user-friendly, and AI-enhanced school management system that improves enrollment and tuition management, strengthens student record management, assists administrative operations, and supports automated reporting for Calvary Christian Academy.

Research Design

The study employed a descriptive-developmental research design supported by mixed-method data collection. The descriptive component focused on examining the existing administrative and record-keeping processes of Calvary Christian Academy (CCA) and identifying the challenges encountered in student records management, enrollment, tuition monitoring, and report generation. The developmental component involved the design and development of the proposed

CCA EduSys: AI-Enhanced School Management System with AI Assistance to address the identified problems and improve the efficiency of the school's administrative operations.

Qualitative data were gathered through an interview with the Registrar Administrator and observations of the existing administrative processes. The observations focused on the actual workflow of enrollment, student records management, tuition monitoring, and report preparation. Quantitative information was also gathered through questionnaires to obtain additional feedback regarding the existing processes and the needs of the intended users. Existing school documents and relevant internet research were also reviewed to support the identification of appropriate system features, requirements, and technologies.

The study involved the school Registrar as the primary respondent, as the position is directly involved in managing student enrollment and records at CCA. Other school personnel, including the principal, teachers, and cashier, were also considered in the observation and information-gathering process to provide additional understanding of the school's administrative procedures. The information gathered from the interview, questionnaires, observations, and document review was analyzed to identify the system requirements and determine the appropriate features and technologies for CCA EduSys.

To support continuous improvement and stakeholder involvement during development, the Agile Model was adopted. The development process follows six phases: Planning, Analysis, Design, Development, Testing, and Deployment. During these phases, the proponents identify the school's needs, analyze the required system features, design the system structure and interface, develop the system modules with AI assistance, test the system for functionality, accuracy, and usability, and prepare it for deployment. User feedback and system performance are considered throughout the development process to allow necessary

improvements and ensure that CCA EduSys meets the administrative needs of Calvary Christian Academy.

Variable Discussion

This study examines several key variables to determine how effective and user-friendly CCA EduSys is in improving the administrative and information management processes of Calvary Christian Academy. The independent variable in this study is CCA EduSys: AI-Enhanced School Management System with AI Assistance, particularly its main features such as centralized student records, enrollment management, tuition payment monitoring, academic record management, automated report generation, and AI-assisted information retrieval. These features serve as the primary elements being introduced in the study, as they are expected to improve how school personnel manage student information and perform administrative tasks. The dependent variables include data redundancy, administrative workload, information retrieval time, report preparation time, and tuition payment-record errors. These factors are expected to improve through the use of the proposed system by reducing repeated data entry, making student and tuition information easier to access, reducing the time required to prepare reports, and improving the accuracy of tuition payment records.

The study considers intervening variables such as Perceived Ease of Use, Perceived Usefulness, and Task-Technology Fit. These variables are based on the Technology Acceptance Model (TAM) and Task-Technology Fit (TTF) Theory and play an important role in determining how users experience and utilize CCA EduSys. Perceived Ease of Use refers to how easily school personnel can learn and operate the system, while Perceived Usefulness refers to how much the system helps users perform their tasks more efficiently. Task-Technology Fit refers to how well the system's features support tasks such as enrollment processing, student record management, tuition monitoring, payment verification, academic record management, and report preparation. Moreover, control variables including the user's assigned role, type of administrative task, and existing school

procedures are also taken into account. These are kept consistent to help ensure that the results of the study are mainly influenced by the proposed system rather than differences in user responsibilities or work processes. These variables provide a clear framework for understanding how CCA EduSys can improve the efficiency, accessibility, and accuracy of school administrative processes while providing users with a system that is useful, easy to use, and suitable for their tasks.

Site, Sampling, and Participants

The study was conducted at Calvary Christian Academy (CCA), located in Sarmiento Townville Subdivision, Barangay Poblacion I, City of San Jose Del Monte, Bulacan, which served as the primary site for data collection, system development, and evaluation. The school was selected because it is the intended environment for the implementation of CCA EduSys: AI-Enhanced School Management System with AI Assistance. The site provided the researchers with direct access to the existing processes involving student records, enrollment, tuition payment monitoring, academic information, and report preparation, allowing the researchers to identify the school's administrative needs and evaluate the proposed system based on its actual operations.

This study utilized purposive sampling to select the Registrar staff as the primary participant because of their direct involvement in student enrollment, records management, and other administrative processes addressed by CCA EduSys. The Registrar staff was selected based on their knowledge, experience, and familiarity with the school's existing procedures, ensuring that the information gathered through the interview was relevant to the identification of system requirements and the development of the proposed system.

The primary participant of the study was therefore the Registrar staff, whose responses, observations, and feedback were used to identify existing problems,

determine system requirements, and provide relevant information for the development and evaluation of CCA EduSys. The participant's direct involvement in the school's administrative operations provided meaningful insights into the existing processes and the requirements needed for the proposed system. This approach ensured that the information gathered was relevant to the actual operations of Calvary Christian Academy and supported the development of a system that addresses the school's identified administrative needs.

Research Instrument

Data collection instruments included structured interviews conducted with administrative and teaching staff to understand current challenges and desired system features. Field observations during peak enrollment and report preparation periods documented workflow inefficiencies and bottlenecks. Secondary research on similar school management systems helped identify best practices and effective technological solutions. Additionally, questionnaires with quantitative measures were distributed to gather empirical data on current system performance and user satisfaction.

Data Gathering Procedures

The data gathering process was conducted in a series of organized steps to ensure that the information collected was accurate, complete, and reliable. Before starting the study, the researchers first asked permission from Calvary Christian Academy and obtained the consent of the Registrar staff, who served as the primary participant. The purpose of the study was clearly explained to ensure that the participant understood the study and their involvement in the data gathering process.

To gather the needed data, the researchers used structured interviews, field observations, and survey questionnaires. The structured interview helped gather detailed information about the existing processes and problems in student records, enrollment, tuition monitoring, and report preparation. Field observations were

conducted to identify workflow difficulties and inefficiencies, while questionnaires were used to gather measurable information regarding the existing processes and the needs of the intended users.

After the development and deployment of CCA EduSys, evaluation questionnaires were administered to assess the quality, usability, user acceptance, and task suitability of the proposed system. The ISO/IEC 25010 Software Quality Model was used to evaluate the quality of the system, while the Technology Acceptance Model (TAM) was used to measure the users' perceived ease of use and perceived usefulness. The Task-Technology Fit (TTF) Theory was also used to determine how well the system's features support the tasks performed by the users, such as student record management, enrollment, tuition monitoring, academic information management, and report preparation.

The questionnaires were based on the participants' actual experience while using CCA EduSys. The researchers provided proper instructions on how to use the system and answer the evaluation questionnaires, while participation remained voluntary throughout the process. Lastly, all collected interview responses, observation results, and questionnaire data were carefully organized and recorded for statistical analysis to determine the system's quality, acceptance, task suitability, and effectiveness in addressing the identified problems of Calvary Christian Academy.

Statistical Treatment

Collected data were analyzed using descriptive statistics. Frequency counts and percentage distributions summarized responses regarding task durations, system usage frequency, and report preparation times. Mean values were calculated to determine average time spent on administrative tasks and typical delays in report generation. These analyses provided a clear understanding of current system limitations and guided the system design process to address key inefficiencies.

Ethical Consideration

The study adhered strictly to ethical standards. Participation by staff was voluntary, with clear explanations of the study's purpose and assurances of confidentiality and anonymity. Respondents were free to withdraw at any time without consequence and were encouraged to ask questions for clarity. All data collected were securely stored and used exclusively for research purposes, ensuring respect and protection for all participants throughout the study.

b. Information Model (Conceptual Design)

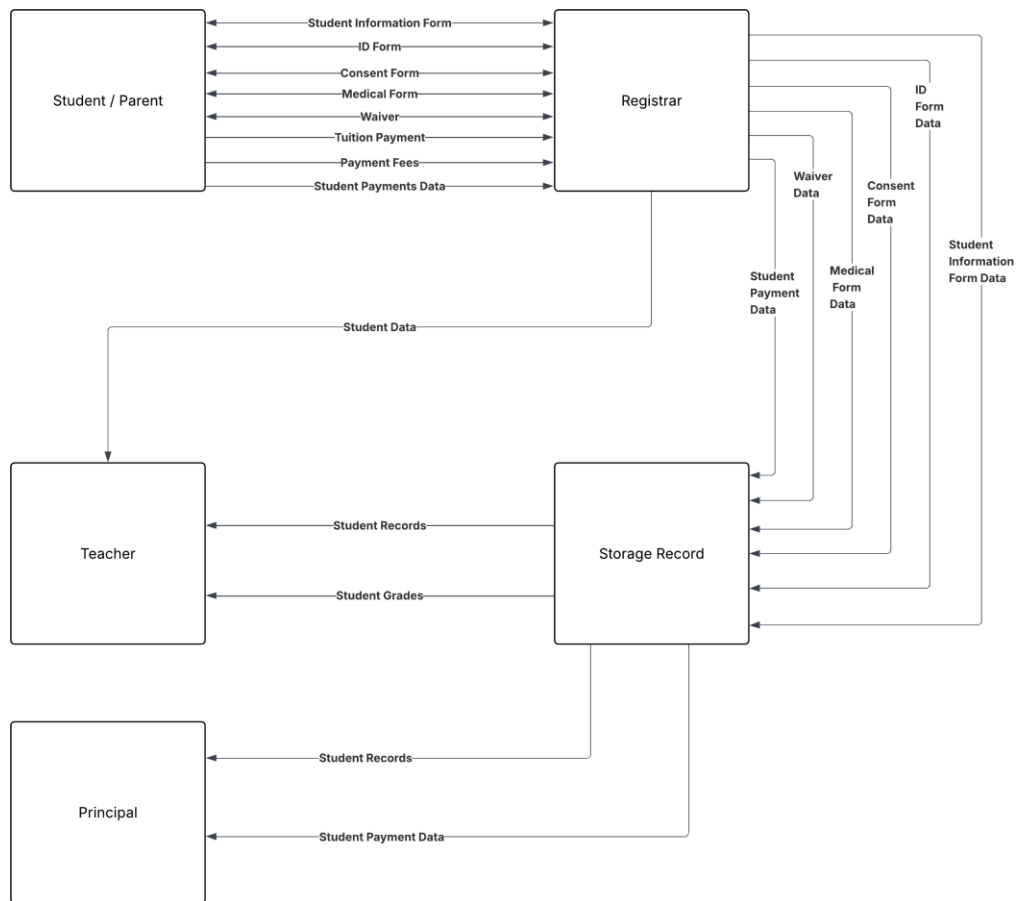


Figure 11. Information Model.

The Information Model illustrates how student-related data flows within the proposed system, highlighting the interaction between Student/Parent, Registrar, Teacher, Principal, and the Storage Record as the central data repository. The process begins with the Student/Parent, who provides documents such as the Student Information Form, ID Form, Consent Form, Medical Form, Waiver, Tuition Payment, Payment Fees, and Student Payments Data. These inputs are received and processed by the Registrar, who collects, organizes, and validates all enrollment and payment-related information before storing them in the system.

The Registrar then transfers structured data—including Student Information Form Data, ID Form Data, Consent Form Data, Medical Form Data, Waiver Data, and Student Payment Data—to the Storage Record, which serves as the centralized database. This ensures records are securely maintained and easily accessible. From this repository, the Teacher retrieves student records and student data and submits student grades, which are stored and updated to support accurate academic monitoring and data consistency.

The Principal accesses the Storage Record to obtain student records and student payment data for decision-making. The model demonstrates a centralized and streamlined data flow, reducing redundancy and manual processing. By consolidating all data into a single Storage Record, the system improves data accuracy, accessibility, efficiency, and real-time coordination among stakeholders.

c. Context Diagram Level 0 (Physical Design)

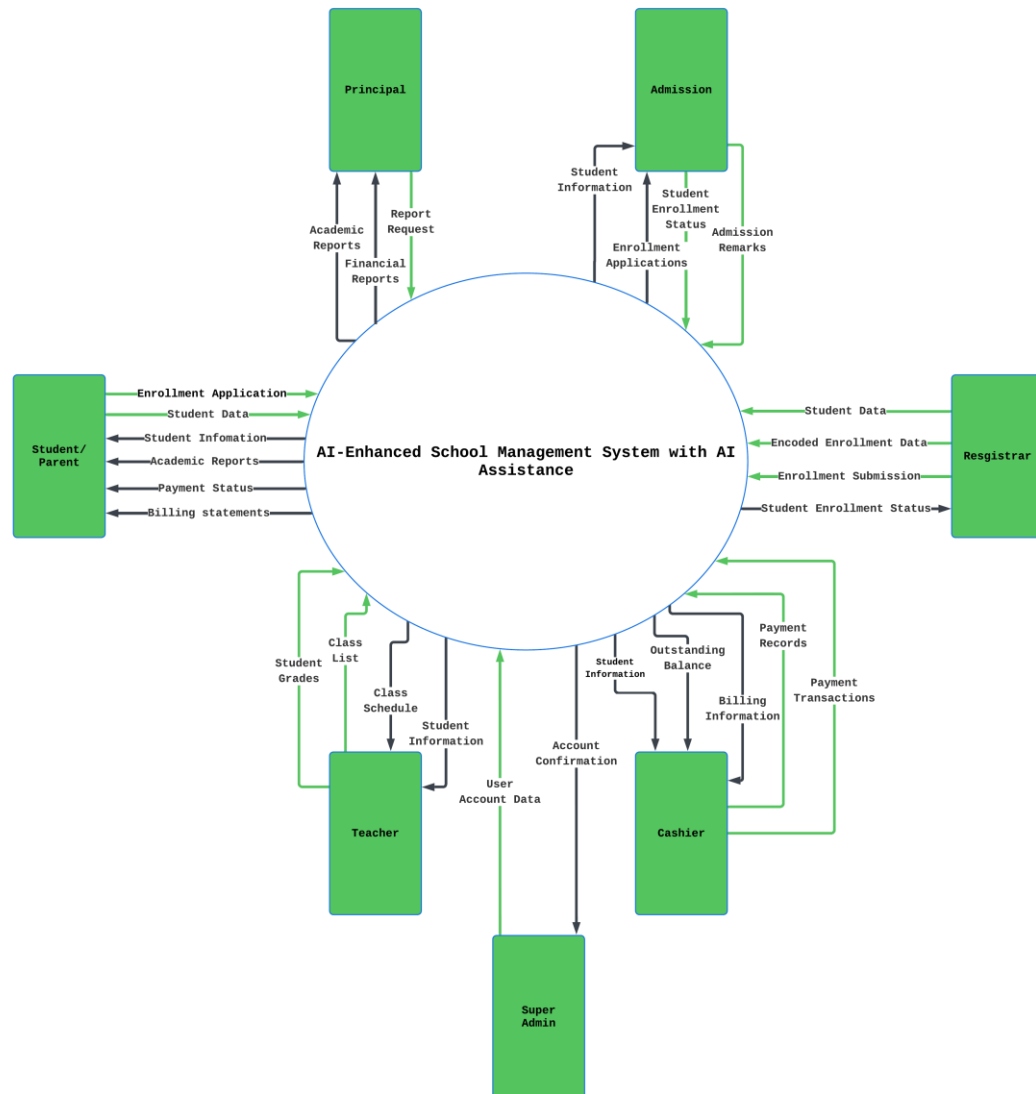


Figure 12. Context Diagram Level 0.

The Context Diagram presents the overall interaction between CCA EduSys: AI-Enhanced School Management System with AI Assistance and its external entities. The system is represented as a single central process that communicates with the different users of Calvary Christian Academy. These external entities include the Student/Parent, Principal, Admission, Registrar, Teacher, Cashier, and

Super Admin, each of whom provides and receives information based on their respective roles and responsibilities within the school.

The Student/Parent provides student-related information, enrollment details, and other necessary data to the system, while receiving student records, academic reports, payment status, and billing statements. The Admission user provides and manages student enrollment information and receives student details and enrollment-related information needed for admission processing. The Registrar provides student and academic records and receives requests and information related to student data, academic records, and enrollment status. The Principal receives academic and financial reports and provides report-related requests or administrative information for monitoring and decision-making.

The Teacher provides grades, attendance, and other academic information and receives student and academic information needed for teaching and record management. The Cashier provides and manages payment and billing information and receives payment records and verification-related information for tuition monitoring. Lastly, the Super Admin provides user account and system administration information and receives user account data and system-related information for managing access and maintaining the system. Through these interactions, CCA EduSys serves as a centralized platform that facilitates the exchange, management, and retrieval of school information among authorized users, while its AI assistance supports information retrieval, inquiries, process guidance, and other user-support functions.

d. Data Flow Diagram Level 1 (Physical Design)

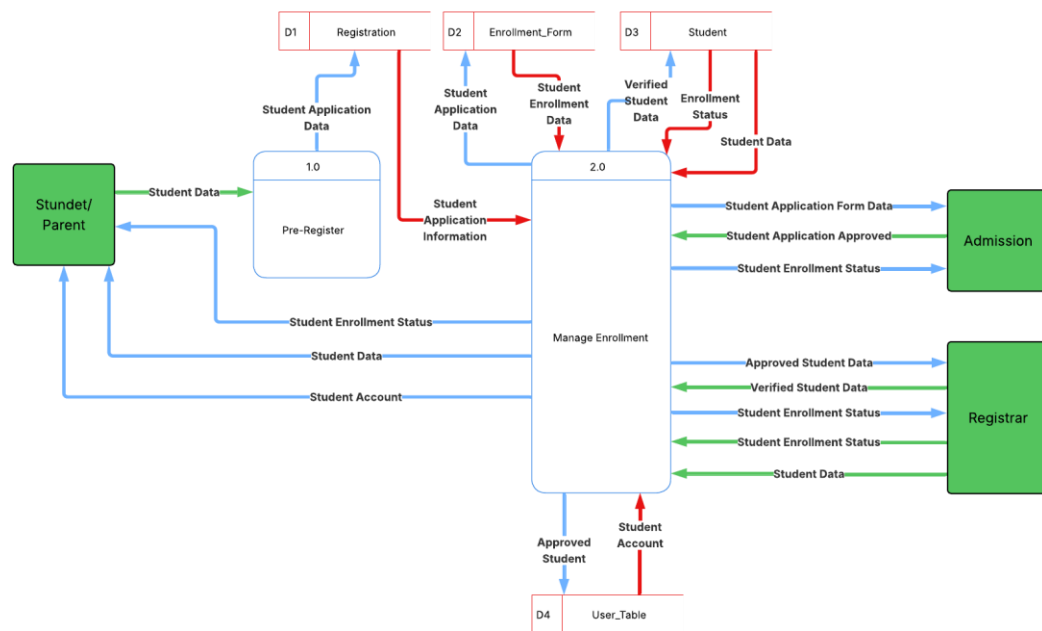


Figure 13.1. Pre-Register & Manage Enrollment Data Flow Diagram.

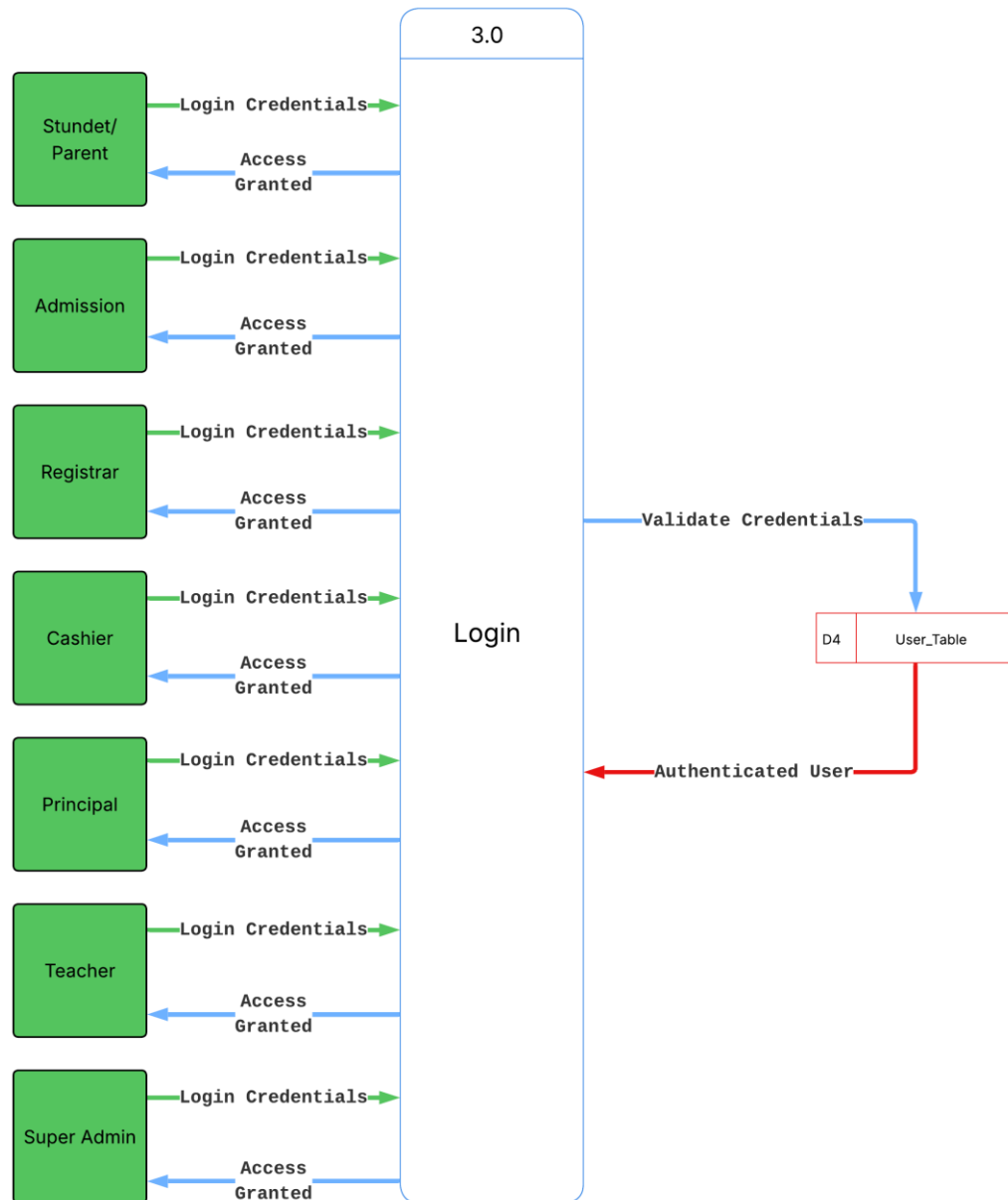


Figure 13.2. Login Data Flow Diagram.

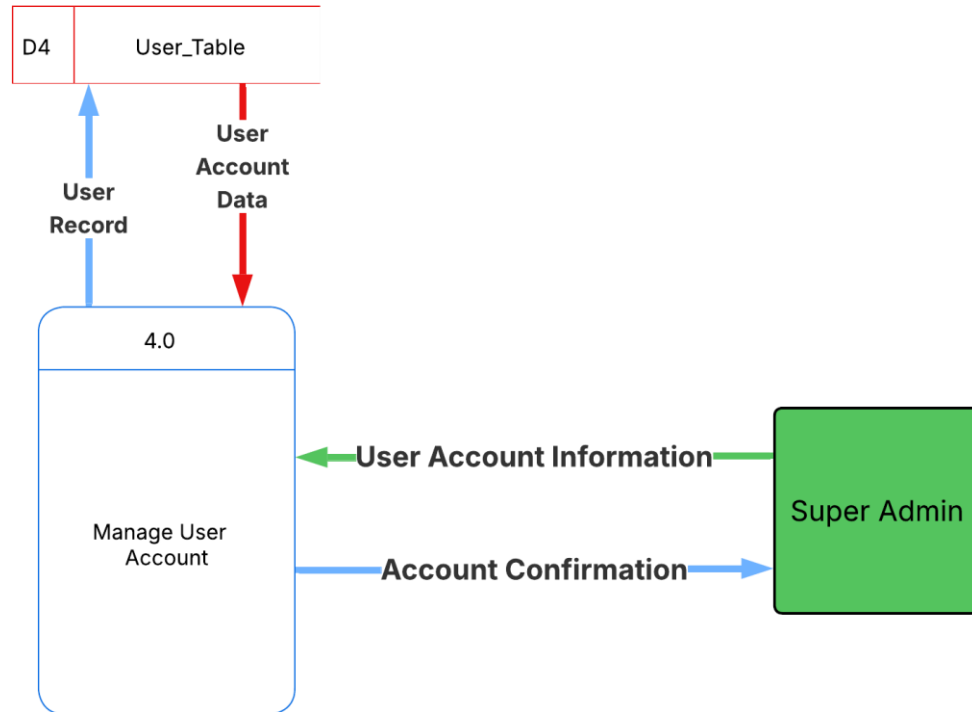


Figure 13.3. *Manage User Account Data Flow Diagram.*

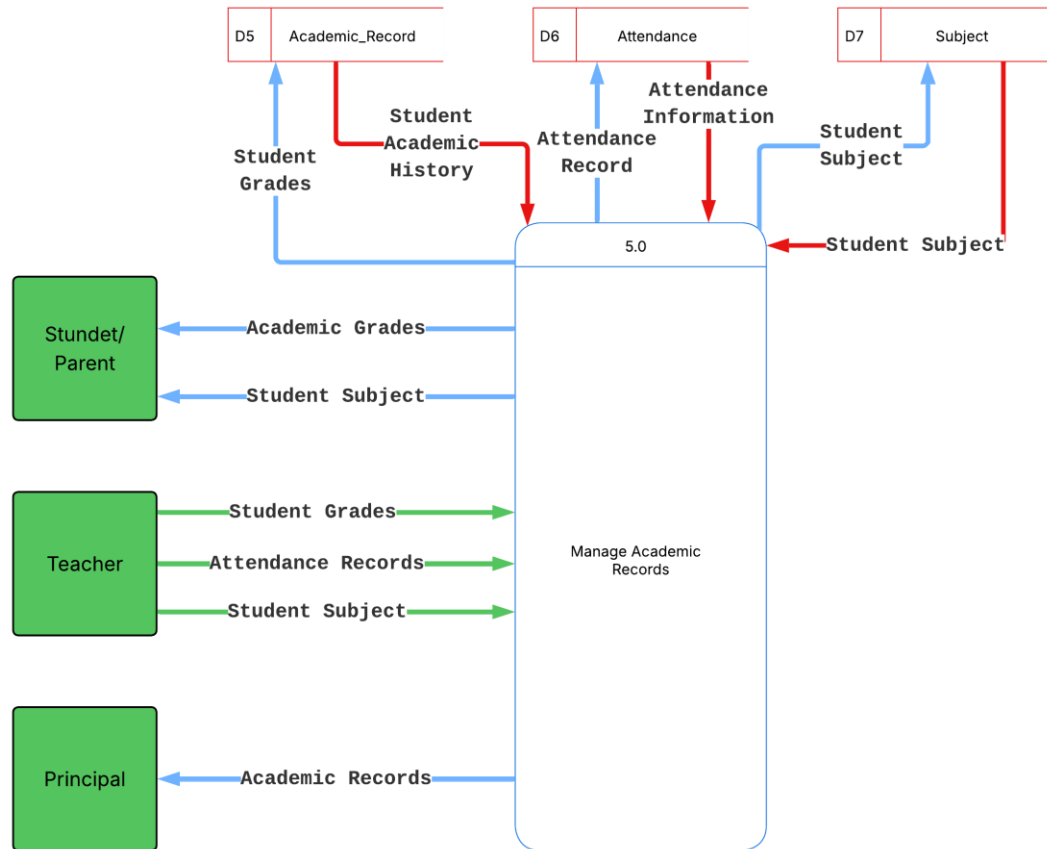


Figure 13.4. *Manage Academic Record Data Flow Diagram.*

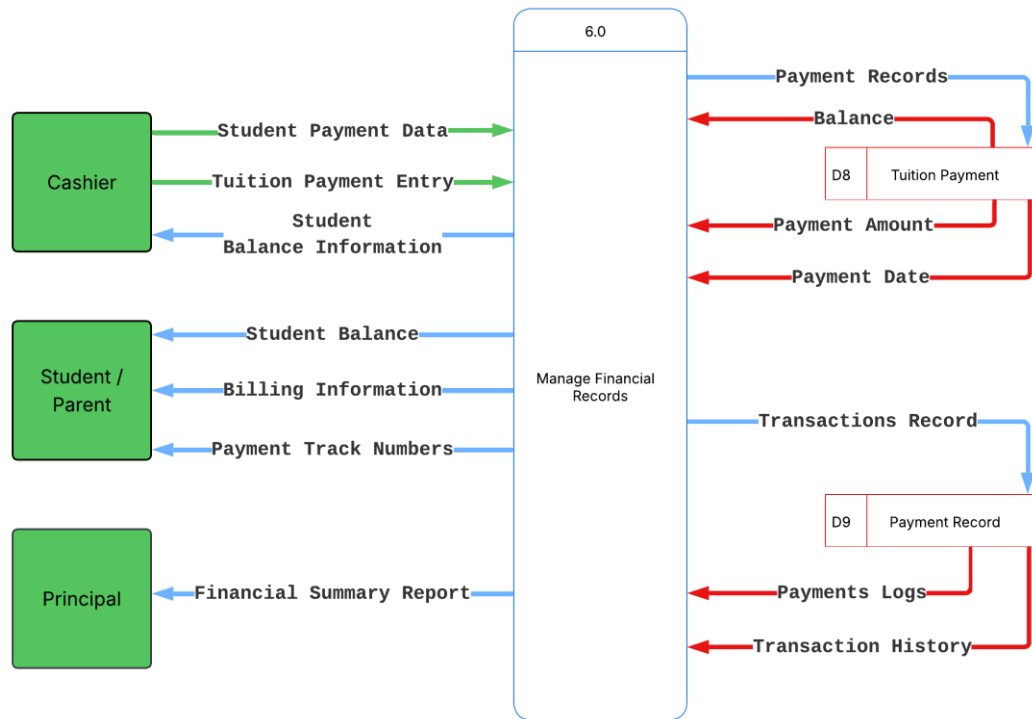


Figure 13.5. *Manage Financial Records Data Flow Diagram*

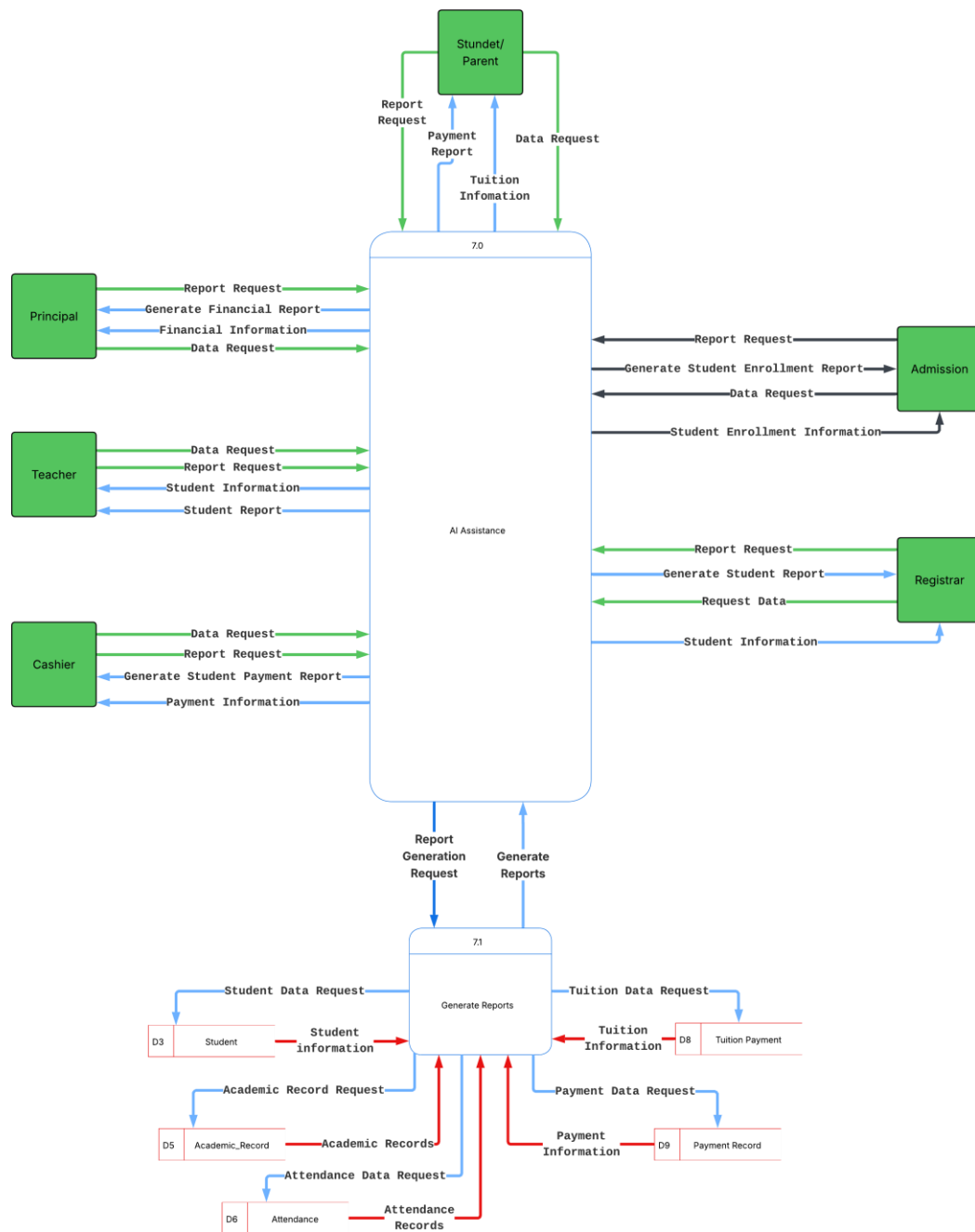


Figure 13.6. *AI Assistance Data Flow Diagram*

e. Entity Relationship Diagram / Database Configuration

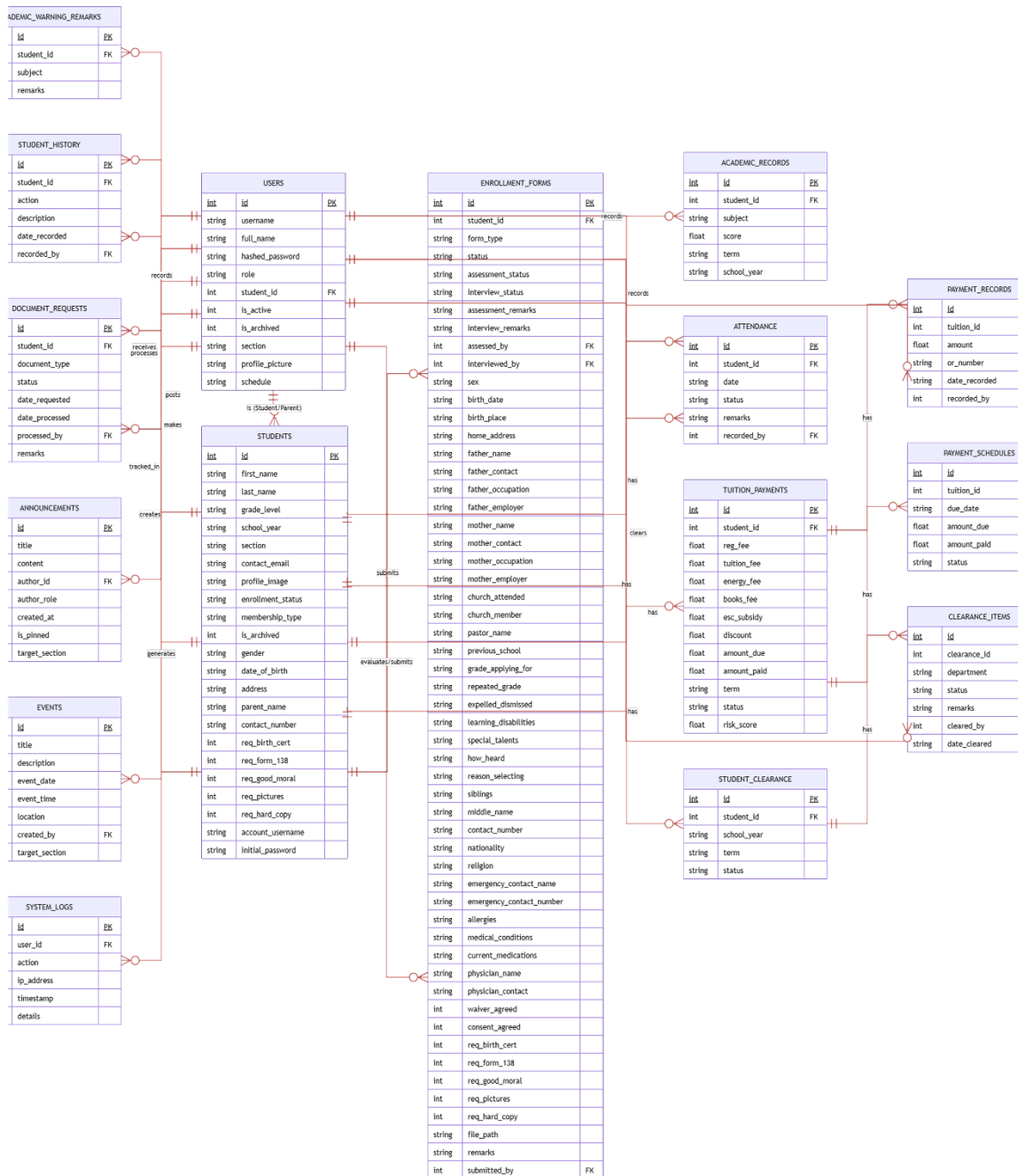


Figure 14. Entity Relationship Diagram.

The proposed AI-Enhanced School Management System for Calvary Christian Academy is designed to centralize and streamline the management of student information, academic records, attendance, tuition payments, and administrative reporting. Based on the Entity-Relationship Diagram (ERD), the system integrates multiple functional modules into a unified database structure, addressing inefficiencies and redundancies in the school's current record-keeping practices. At the core of the system is the User entity, which manages access through role-based accounts such as IT Support, Registrar, Teacher, and Principal, ensuring secure and controlled data access while maintaining data integrity. The Teacher entity is linked to the User table and associated with the Subject entity, enabling teachers to manage assigned subjects effectively.

The Student entity serves as the central component where all student-related data is stored and interconnected with key modules, including Enrollment_Form, Academic_Record, Attendance, Risk_Assessment, Error_Log, and Tuition_Payment, ensuring that data is recorded once and reused across the system. This structure directly addresses repeated data entry and supports the AI Auto-Fill Enrollment feature. The Enrollment_Form entity digitizes the enrollment process by storing forms, school year data, and documents, reducing manual paperwork, while the Academic_Record tracks student performance per subject and quarter for efficient grade monitoring and automated report generation. The Attendance entity records student presence along with the responsible user, improving accountability and real-time monitoring, while the Risk_Assessment entity introduces an AI-driven component that stores predicted risk levels, reasons, and dates to enable early intervention for at-risk students.

To maintain data accuracy, the Error_Log entity records inconsistencies such as duplicating or missing data, supporting the AI Error Detection feature. Financial management is handled through the Tuition_Payment, Payment_Record, and Receipt entities, where tuition details, transaction logs, and proof of payment are stored, improving accuracy, traceability, and efficiency in tuition tracking while

reducing reliance on physical receipts. The Report entity enables the generation of academic and financial summaries linked to system users, supporting the AI Report Generator feature and significantly reducing report preparation time. Overall, the ERD presents a centralized and well-structured database system that integrates all major school operations, minimizes redundancy, enhances data accuracy, reduces administrative workload, and improves decision-making through AI-supported features such as auto-fill enrollment, error detection, risk assessment, and automated reporting.

f. Flowchart

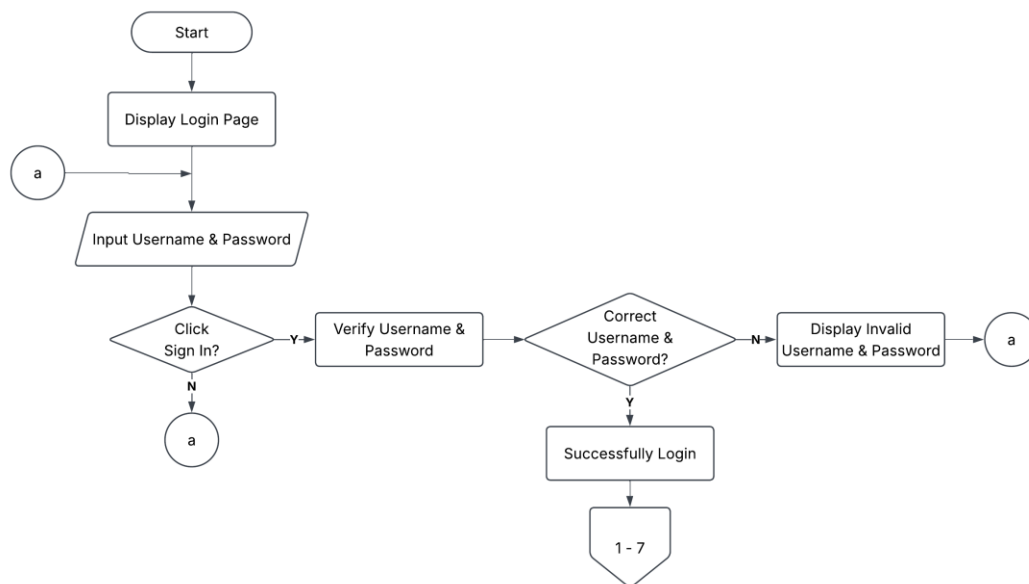


Figure 15.1. Login Page Flowchart.

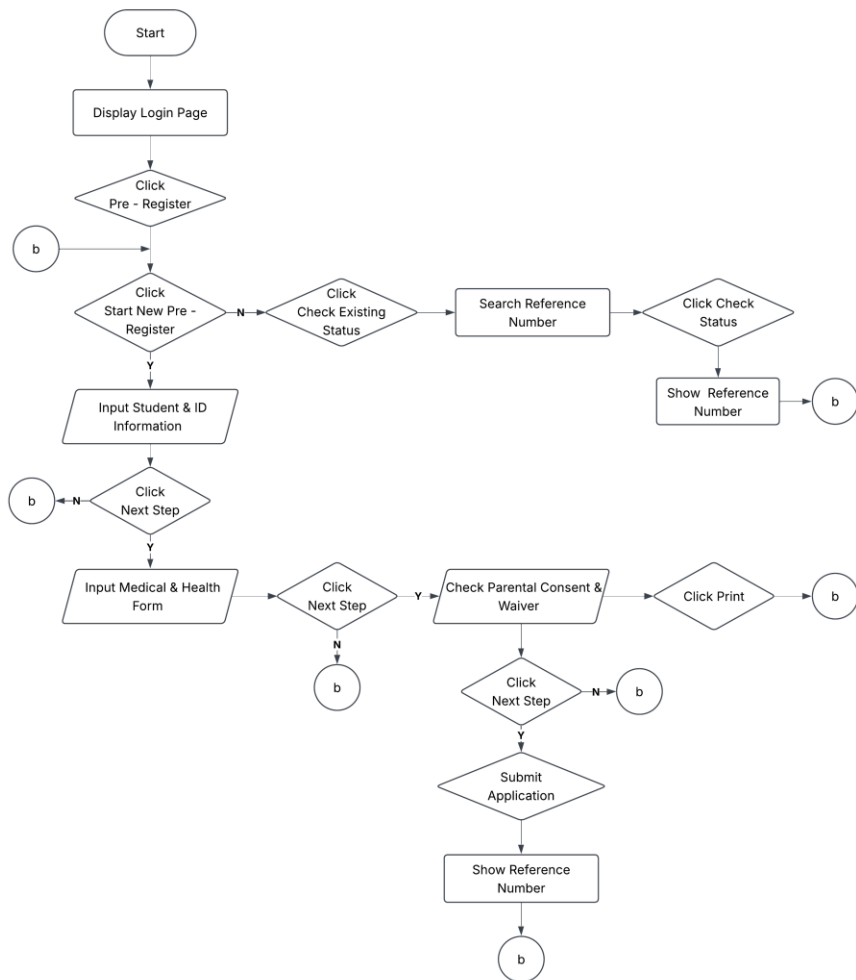


Figure 15.2. Pre-Register Page Flowchart.

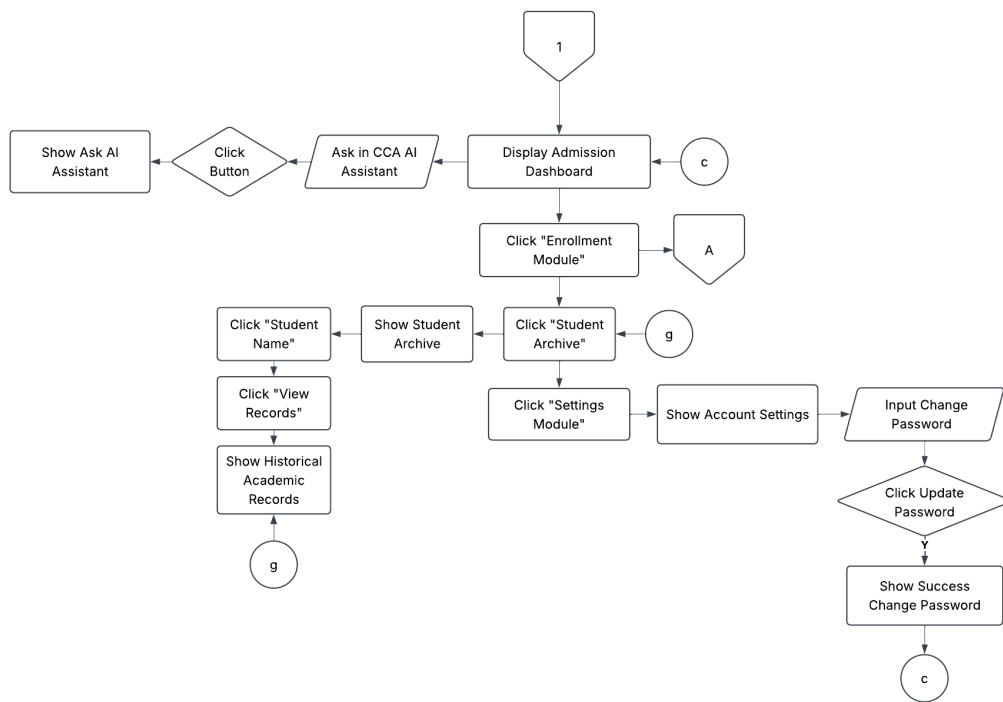


Figure 15.3. Admission Account Flowchart.

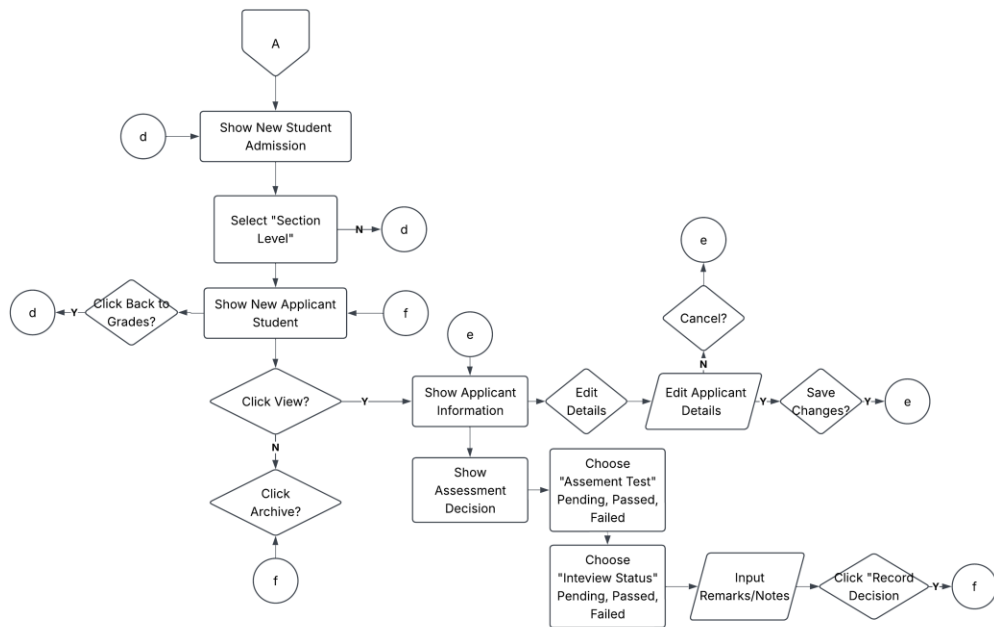


Figure 15.4. Admission Enrollment Flowchart.

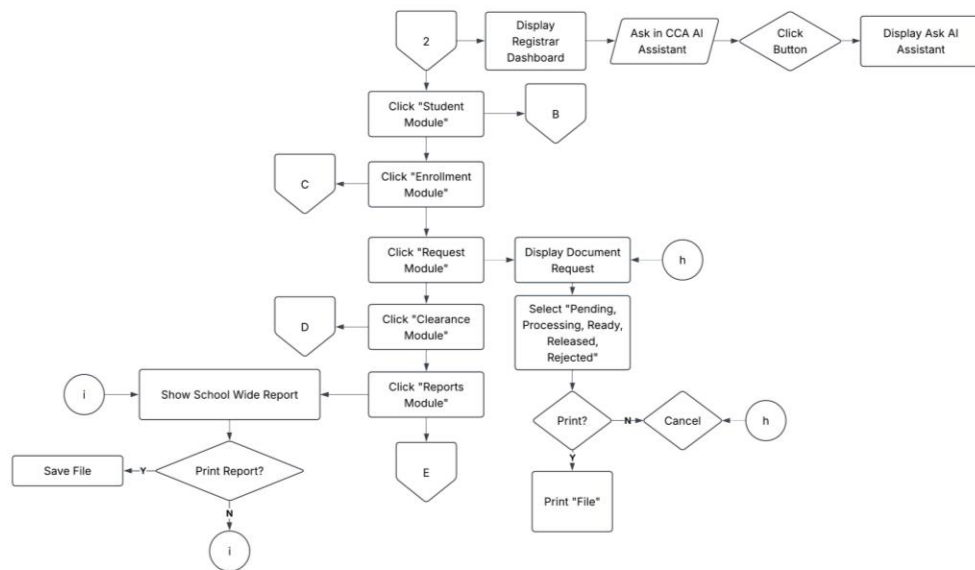


Figure 15.5. Registrar Account Flowchart.

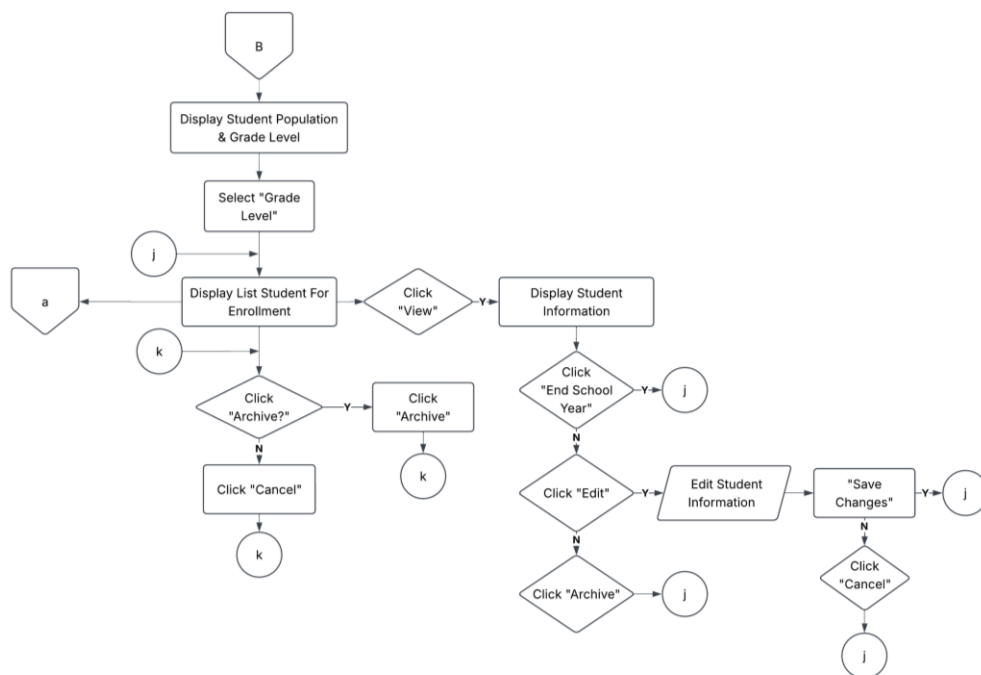


Figure 15.6. Registrar Student Module Flowchart.

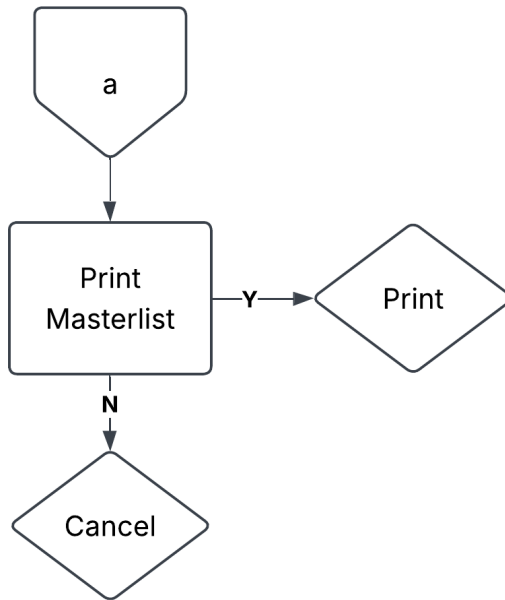


Figure 15.7. Registrar Print Flowchart.

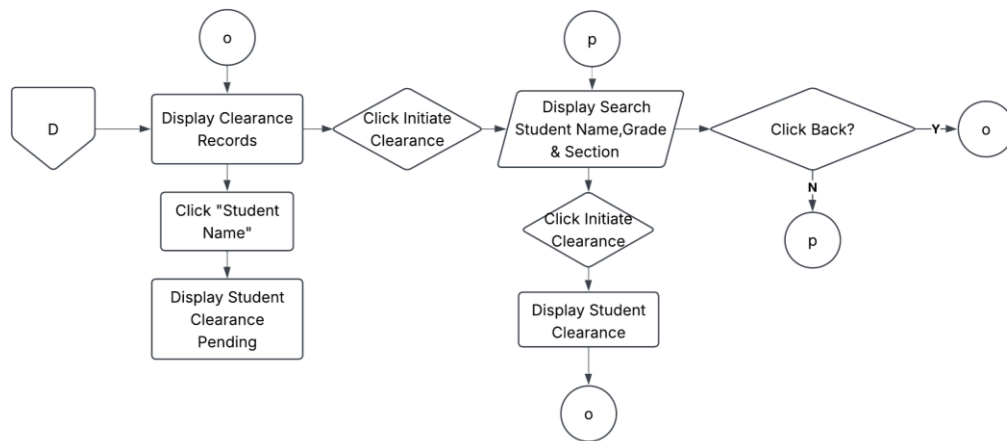


Figure 15.8. Registrar Clearance Module Flowchart.

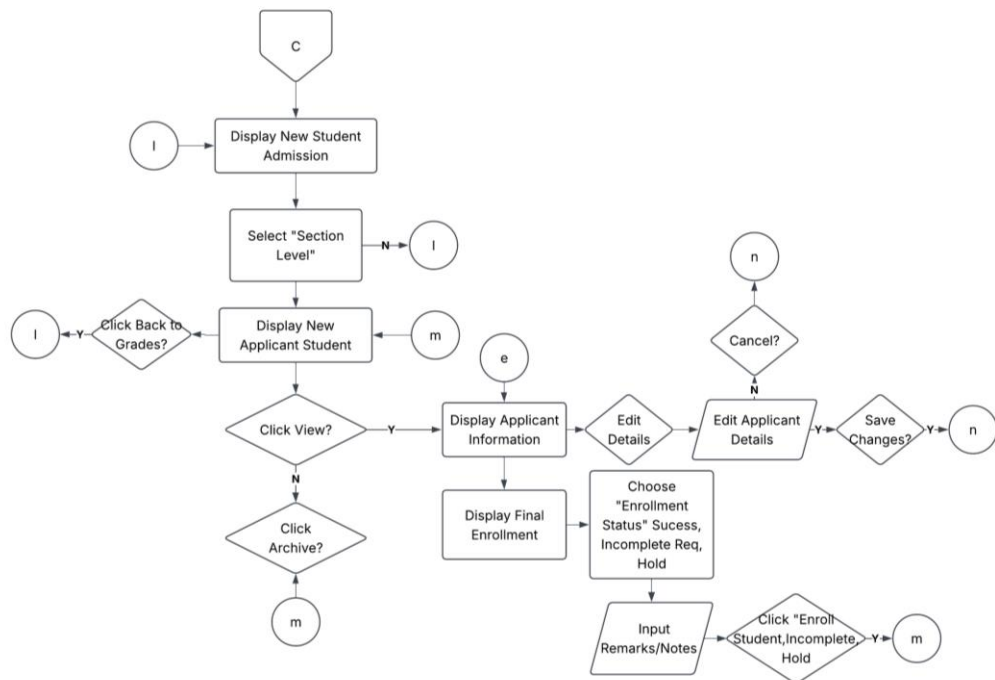


Figure 15.9. Registrar Enrollment Module Flowchart.

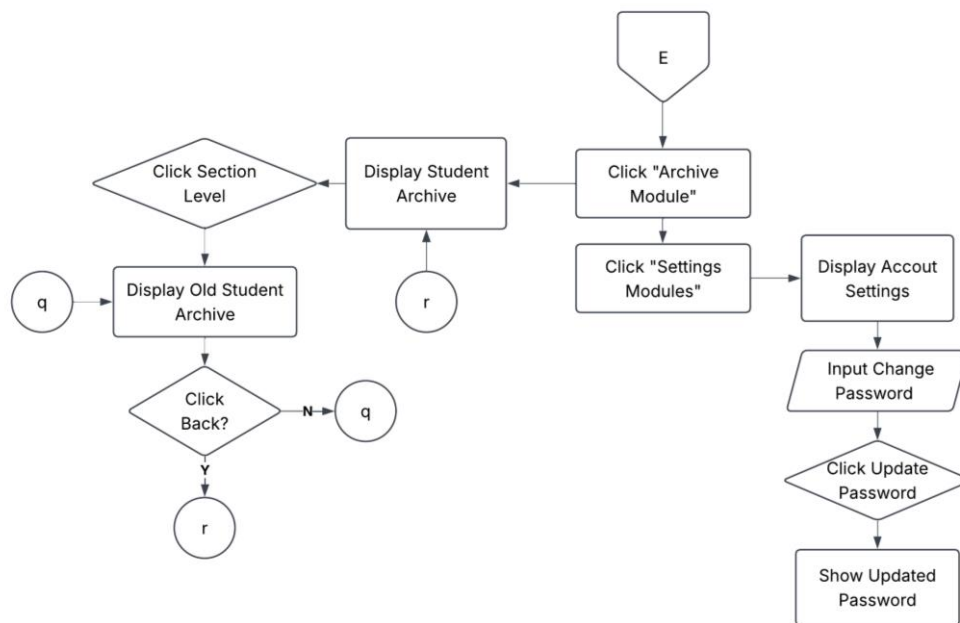


Figure 15.10. Registrar Archive Module Flowchart.

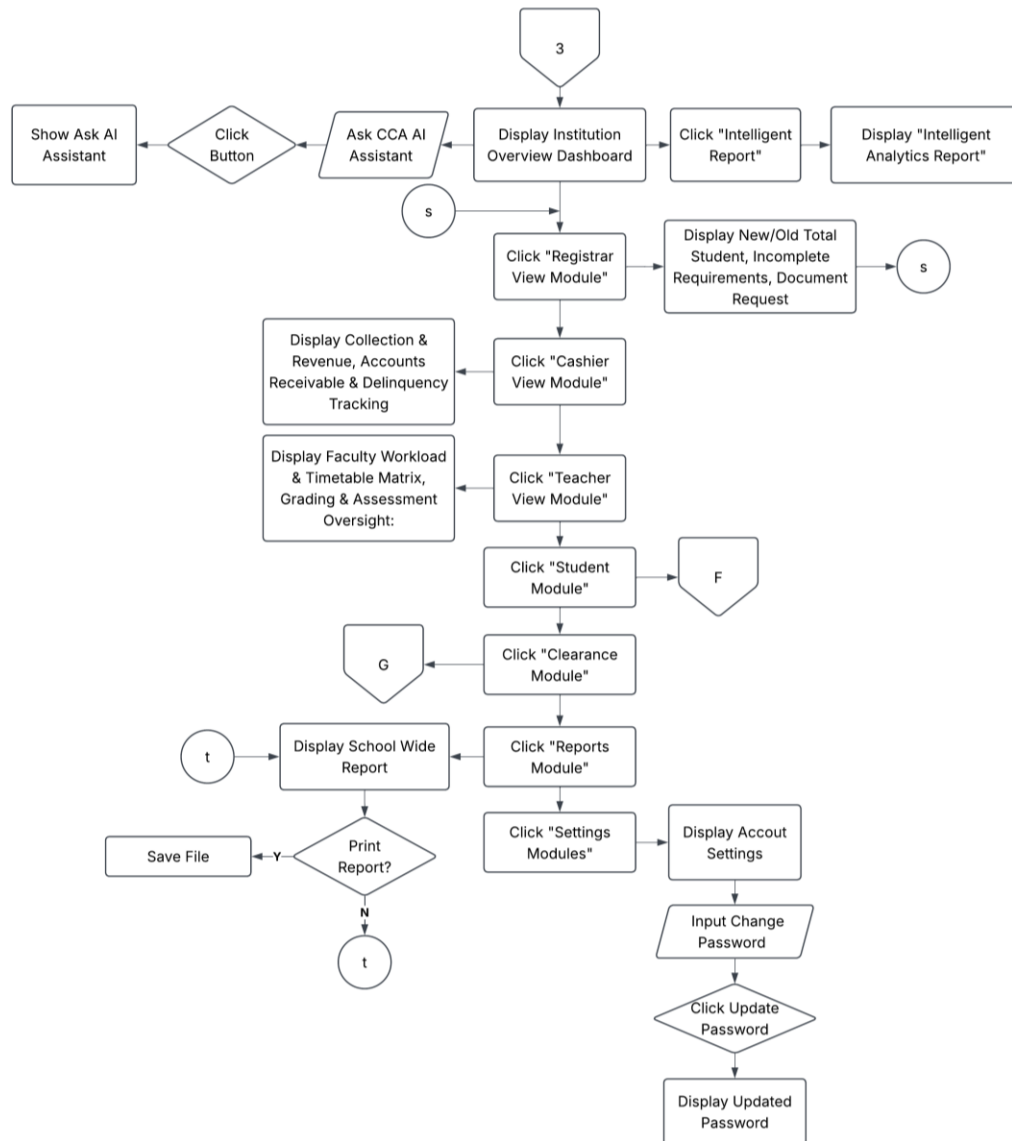


Figure 15.11. *Principal Account Flowchart.*

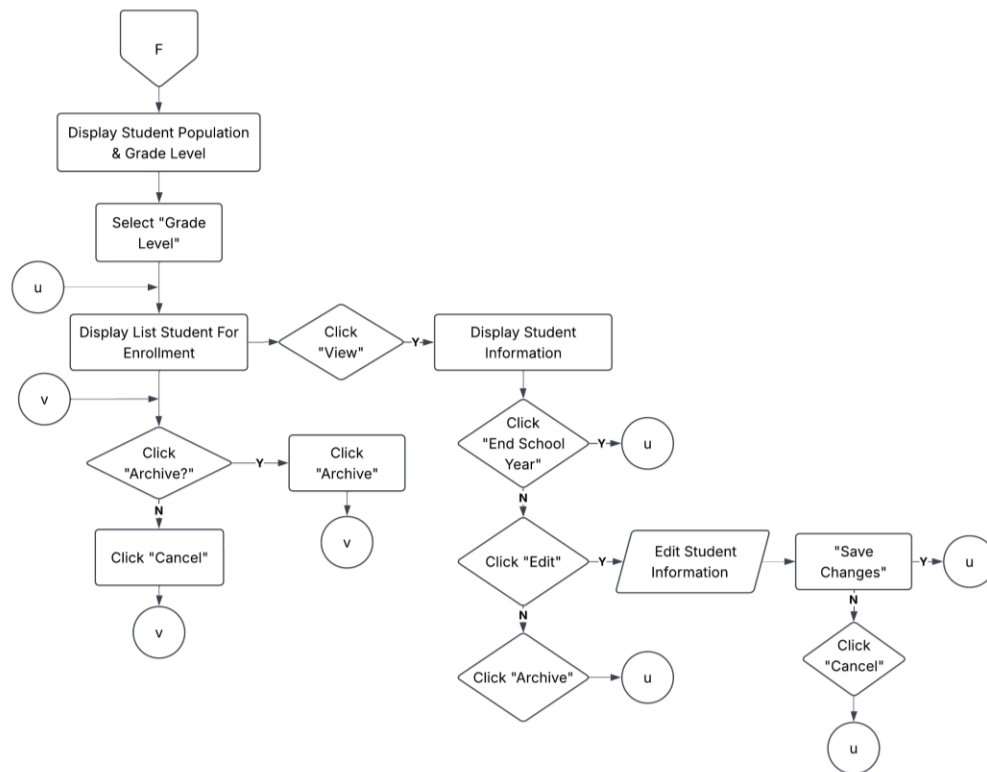


Figure 15.12. Principal Student Module Flowchart.

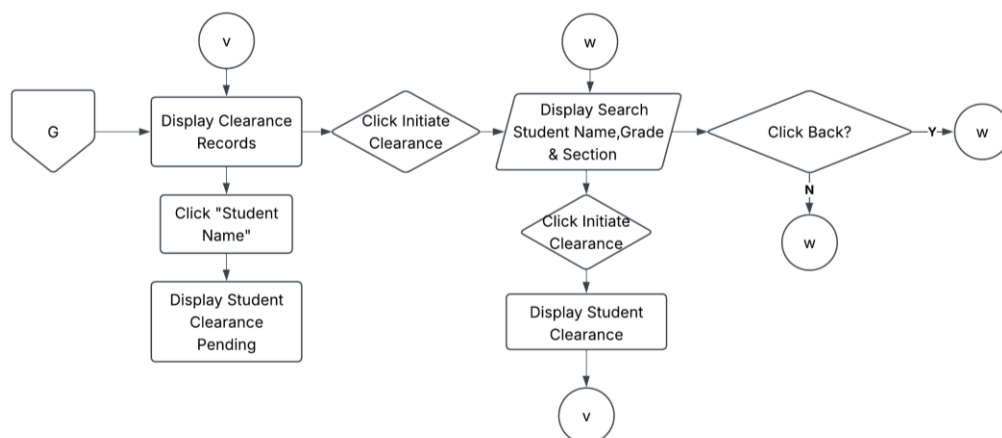


Figure 15.13. Principal Clearance Module Flowchart.

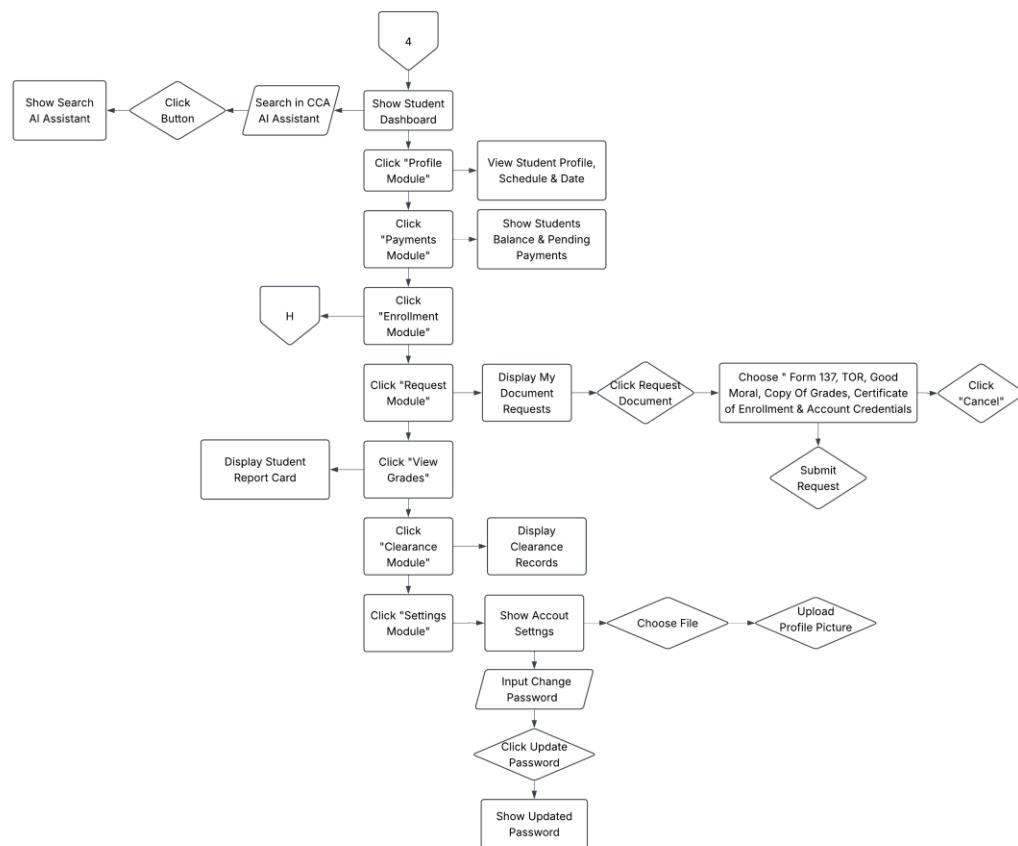


Figure 15.14. Student Account Flowchart.

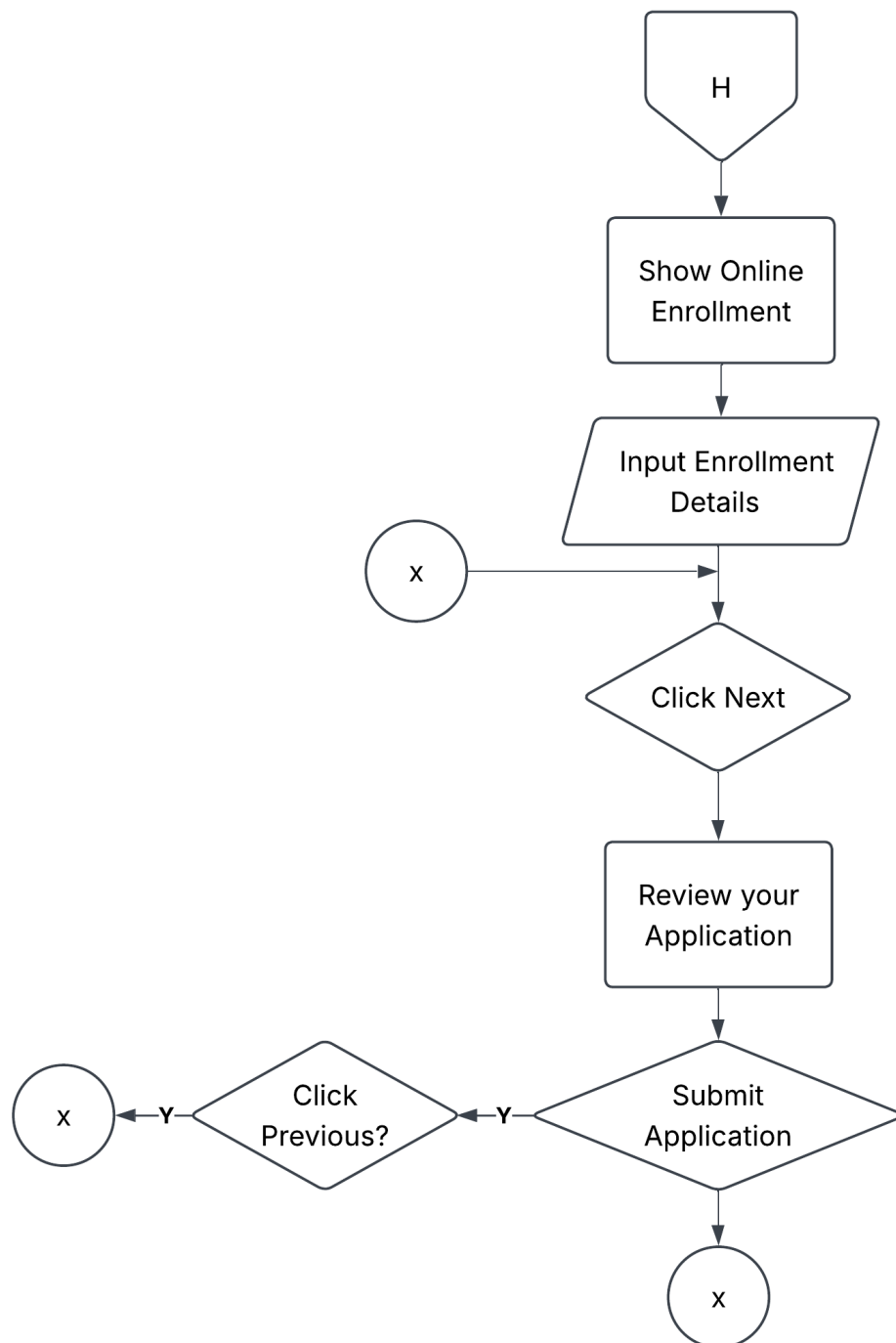


Figure 15.15. *Student Enrollment Module Flowchart.*

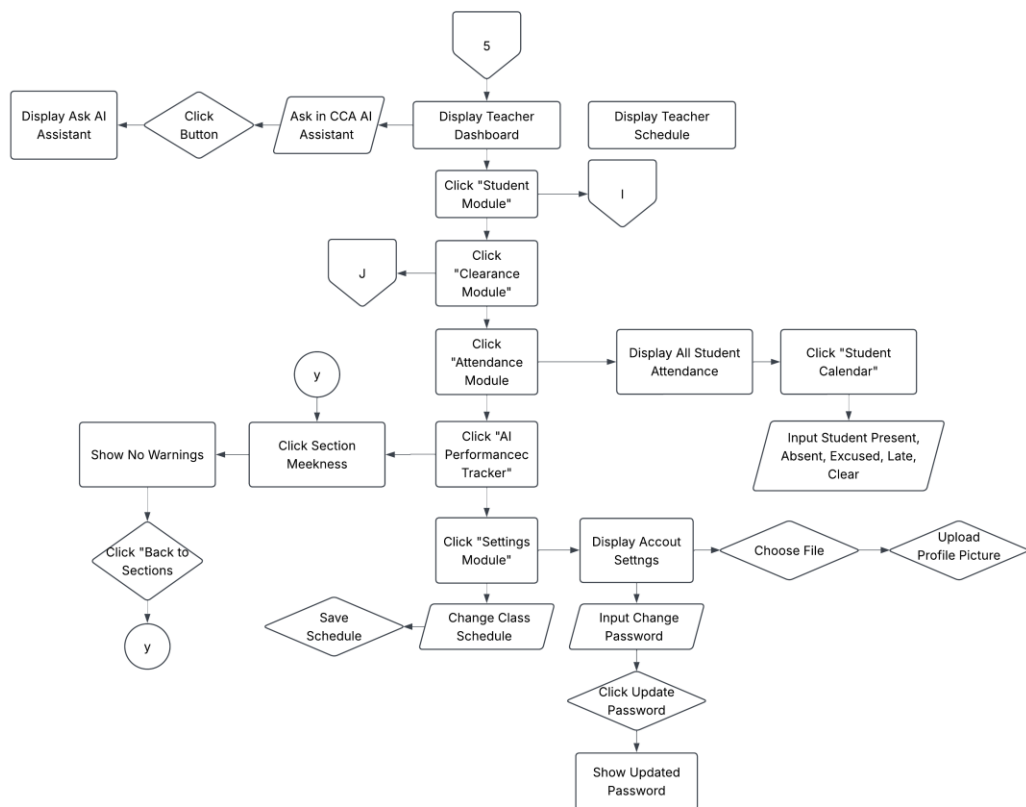


Figure 15.16. Teacher Account Flowchart.

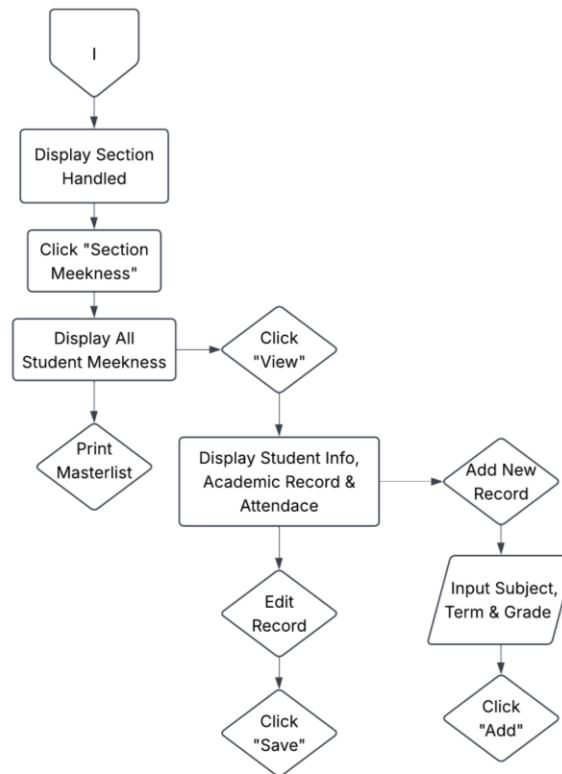


Figure 15.17. Teacher Student Module Flowchart.

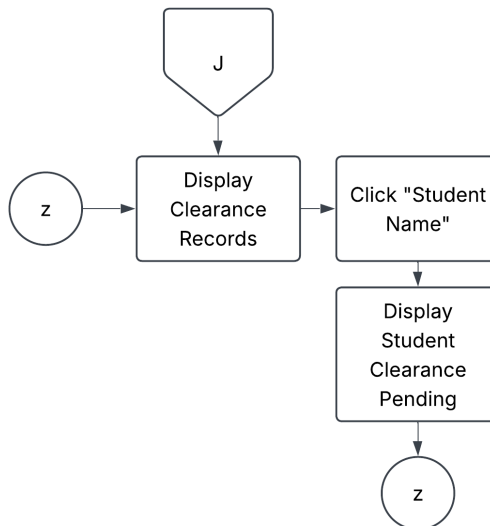


Figure 15.18. Teacher Clearance Module Flowchart.

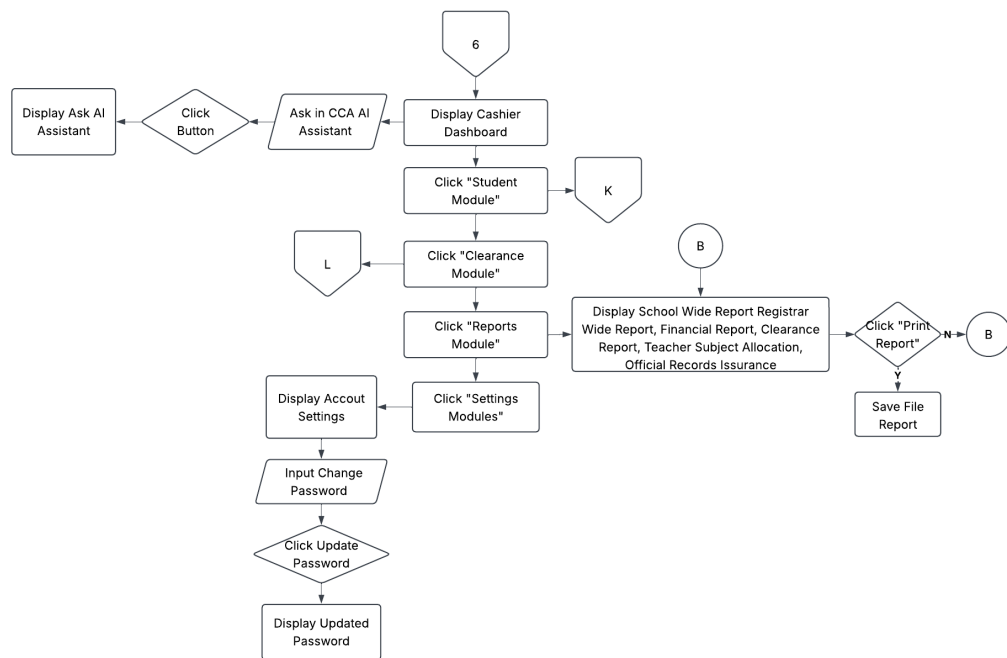


Figure 15.19. *Cashier Account Flowchart.*

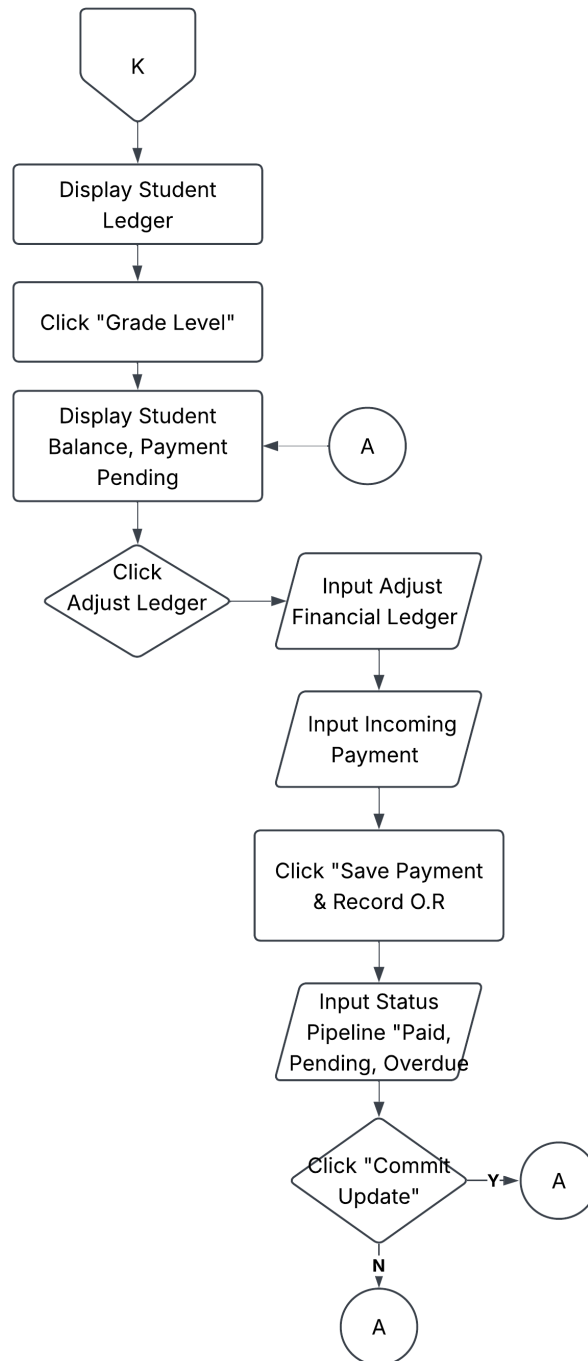


Figure 15.20. *Cashier Student Module Flowchart.*

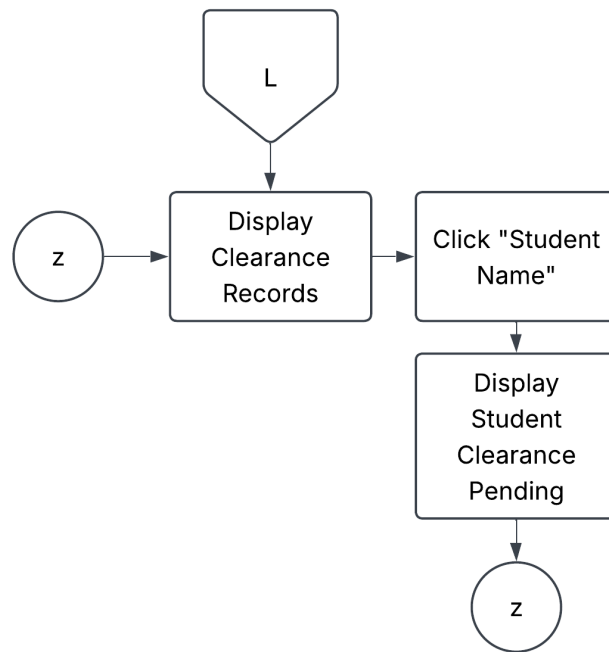


Figure 15.21. *Cashier Clearance Module Flowchart.*

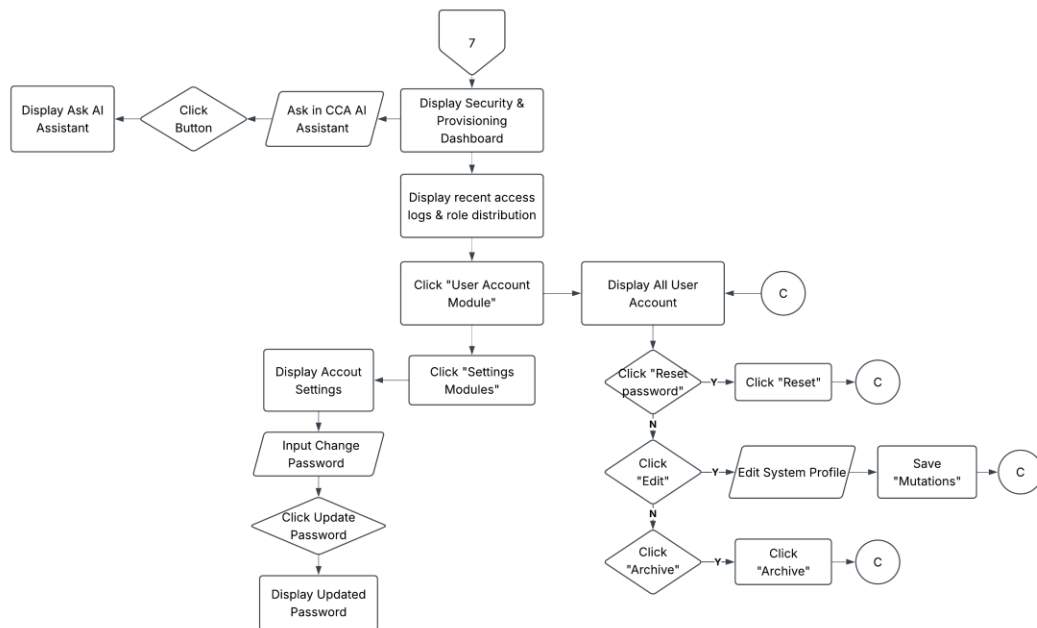


Figure 15.22. *Super Admin Account Flowchart.*

g. Architecture Design

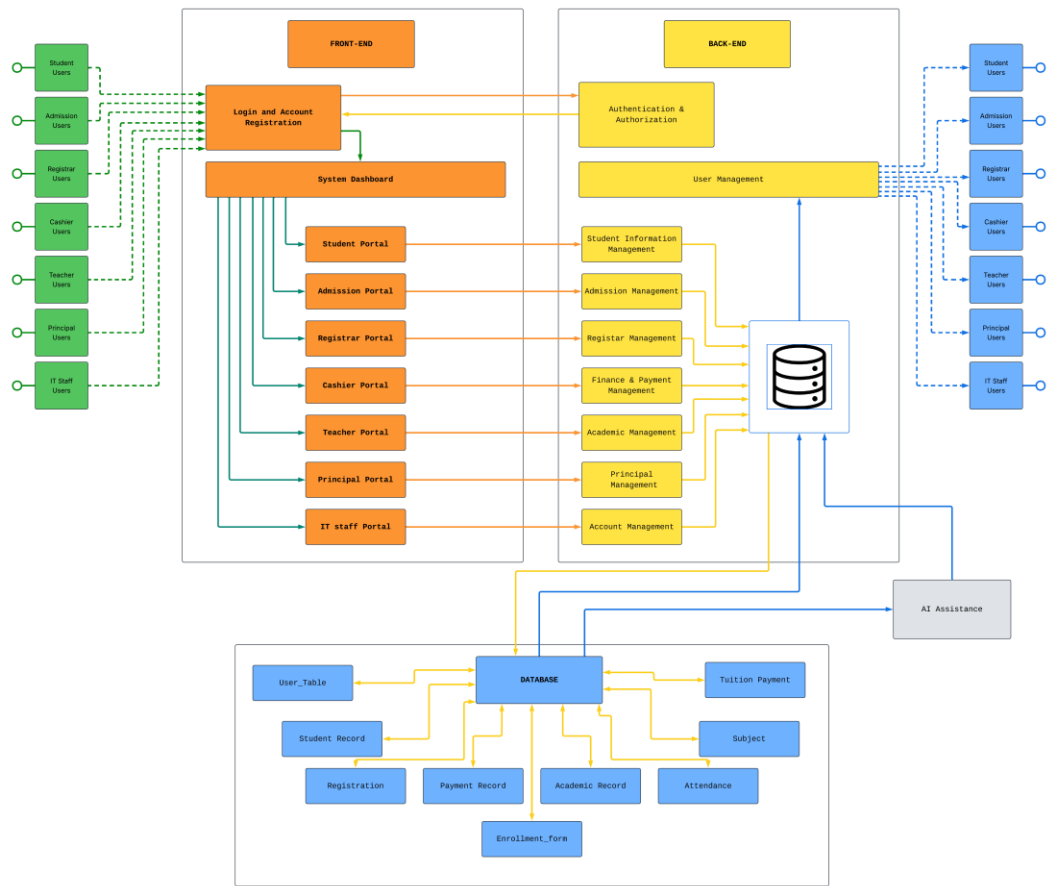


Figure 16. System Architecture Design.

The architecture design of CCA EduSys: AI-Enhanced School Management System with AI Assistance illustrates the overall structure and interaction between the system's users, front-end, back-end, database, and AI assistance. The architecture is organized into several interconnected layers to ensure that users can access the appropriate functions while maintaining centralized and secure management of school information.

The user layer consists of the Student, Admission, Registrar, Cashier, Teacher, Principal, and IT Staff users. These users access the system through their respective accounts and authorized portals. The front-end layer provides the user

interface of CCA EduSys, including the Login and Account Registration, System Dashboard, Student Portal, Admission Portal, Registrar Portal, Cashier Portal, Teacher Portal, Principal Portal, and IT Staff Portal. These interfaces allow users to perform tasks based on their assigned roles and permissions.

The back-end layer handles the main system operations and business logic. It includes Authentication and Authorization for verifying user identities and controlling access, User Management for managing accounts and roles, and the different management modules for Student Information, Admission, Registrar, Finance and Payment, Academic, Principal, and Account Management. The back-end communicates with the database to store, retrieve, and update information required by the different system modules.

The database layer serves as the centralized storage of CCA EduSys. It contains important records such as User Table, Student Record, Registration, Payment Record, Academic Record, Enrollment Form, Tuition Payment, Subject, and Attendance. Centralizing these records allows authorized modules and users to access consistent and organized information while reducing redundant data entry and improving information retrieval. The architecture also integrates an AI Assistance component that works with the system's information and management functions to support intelligent search, natural language inquiries, information retrieval, process guidance, report summarization, and other user-assistance functions. Through the interaction of these components, the architecture provides a centralized, role-based, and AI-assisted environment for managing the academic and administrative operations of Calvary Christian Academy.

h. Algorithm Design (UI/UX)

Figure 17.1. Login Page.

Figure 17.2. Student Info Pre-Registration Page.

Figure 17.3. Medical History Pre-Registration Page.

Figure 17.4. Waiver for face-to-face Classes Pre-Registration Page.

Figure 17.5. Consent Form Pre-Registration Page.

Figure 17.6. Pre-Registration Submitted Page.

Figure 17.7. Admission Dashboard Page.

Figure 17.8. Admission Enrollment Page.

NAME	ADMISSION	ADMISSION	ADMISSION	ADMISSION
1000	Admission	1000	Admission	1000
1001	Admission	1001	Admission	1001
1002	Admission	1002	Admission	1002
1003	Admission	1003	Admission	1003
1004	Admission	1004	Admission	1004
1005	Admission	1005	Admission	1005

Figure 17.9. Admission List of Students Page.

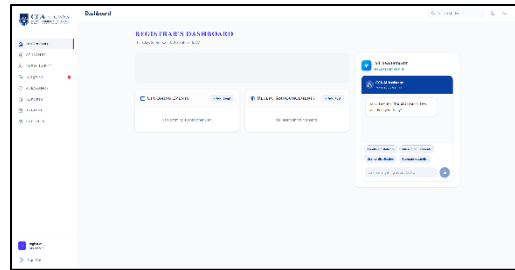


Figure 17.10. Registrar Dashboard Page.



Figure 17.11. Registrar Students Page.

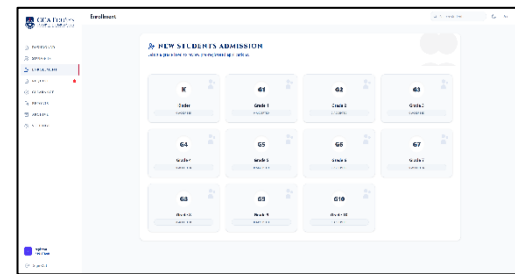


Figure 17.12. Registrar Enrollment Page.

NAME	REQUEST	REQUEST	REQUEST	REQUEST
1000	Request	1000	Request	1000
1001	Request	1001	Request	1001
1002	Request	1002	Request	1002
1003	Request	1003	Request	1003
1004	Request	1004	Request	1004
1005	Request	1005	Request	1005

Figure 17.13. Registrar Request Page.

NAME	REPORT	REPORT	REPORT	REPORT
1000	Report	1000	Report	1000
1001	Report	1001	Report	1001
1002	Report	1002	Report	1002
1003	Report	1003	Report	1003
1004	Report	1004	Report	1004
1005	Report	1005	Report	1005

Figure 17.14. Registrar Reports Page.

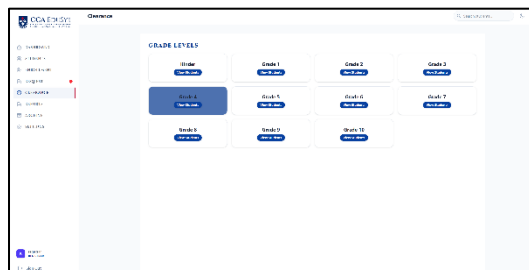


Figure 17.15. Registrar Clearance Page.

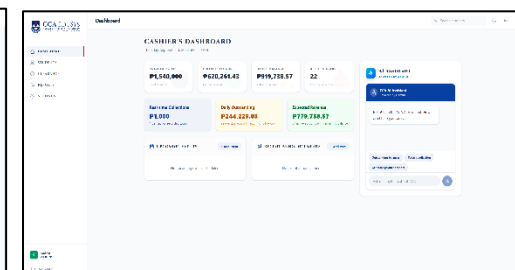


Figure 17.16. Cashier Dashboard Page.

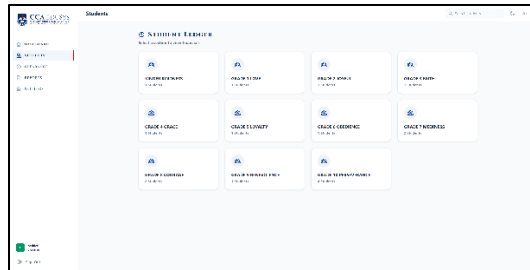


Figure 17.17. Cashier Students Page.

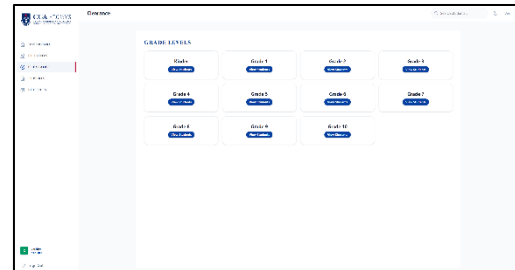


Figure 17.18. Cashier Clearance Page.

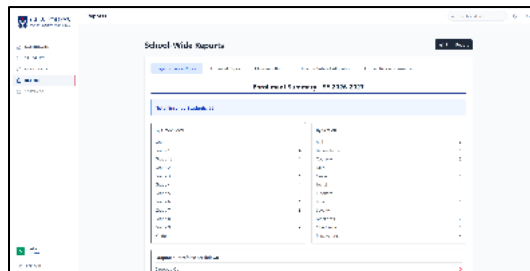


Figure 17.19. Cashier Reports Page.

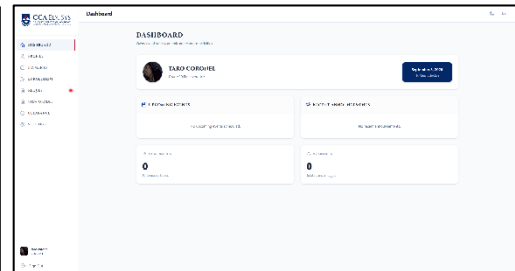


Figure 17.20. Student Dashboard Page.

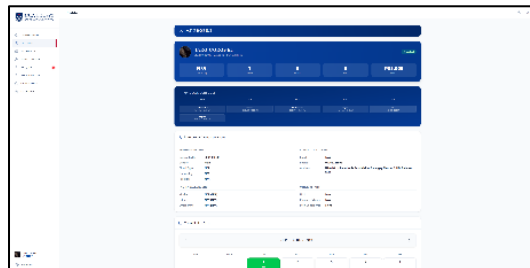


Figure 17.21. Student Profile Page.

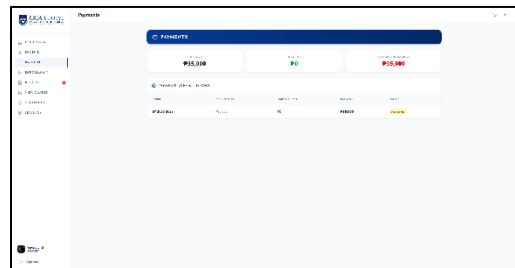


Figure 17.22. Student Payments Page.

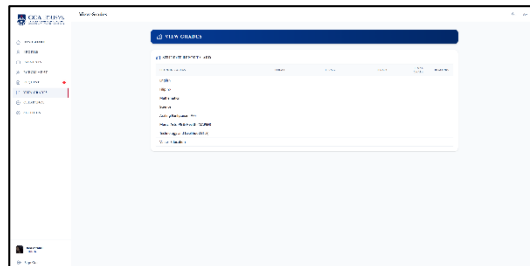


Figure 17.23. Student View Grade Page.

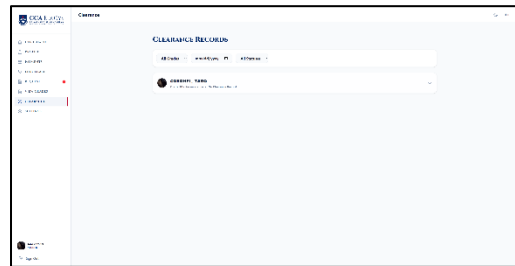


Figure 17.24. Student Clearance Page.

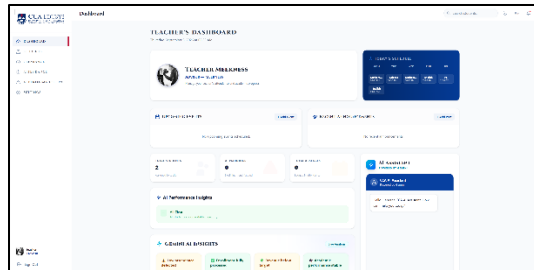


Figure 17.25. Teacher Dashboard Page.

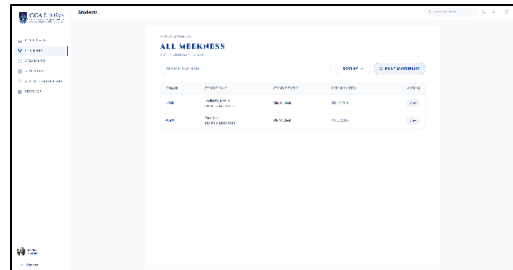


Figure 17.26. Teacher Students Page.

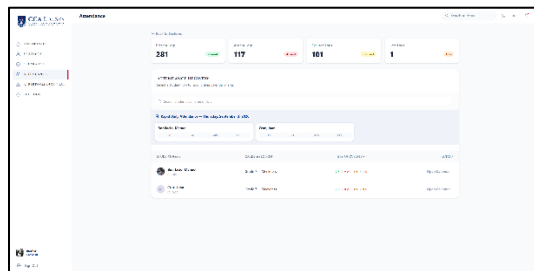


Figure 17.27. Teacher Attendance Page.

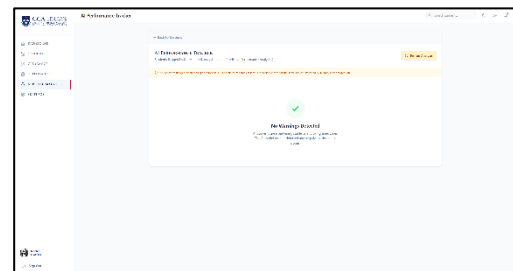


Figure 17.28. Teacher AI Performance Tracker Page.

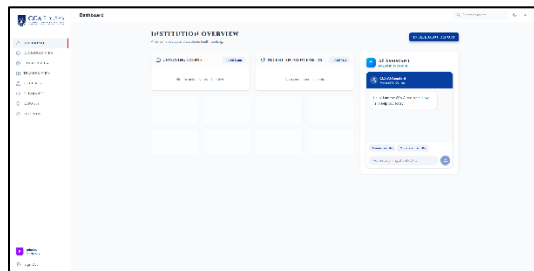


Figure 17.29. Principal Dashboard Page.

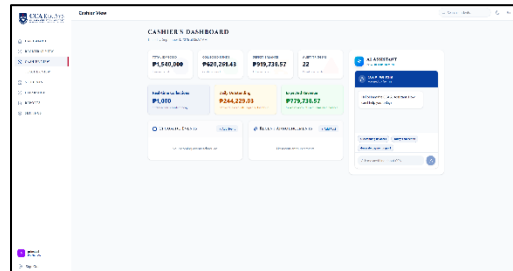


Figure 17.30. Principal Cashier Reports Page.

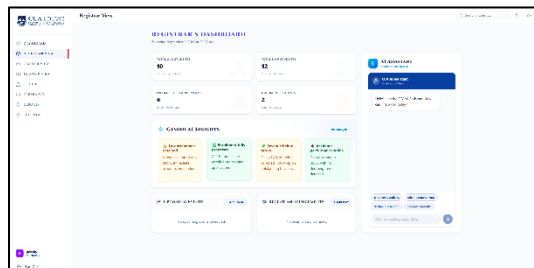


Figure 17.31. Principal Registrar Reports Page.

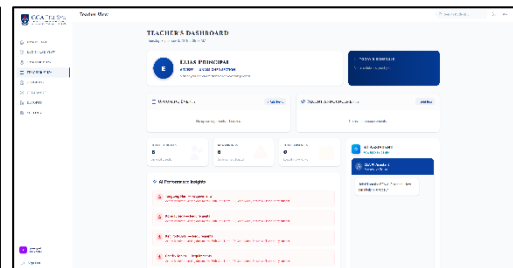


Figure 17.32. Principal Teacher Reports Page.



Figure 17.33. Principal Student Reports Page.

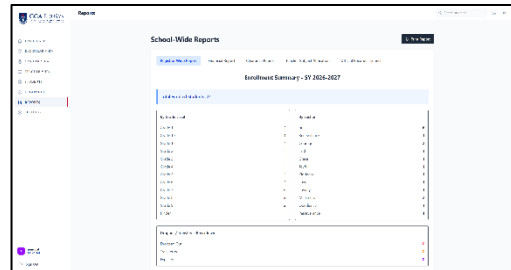


Figure 17.34. Principal School-Wide Reports Page.

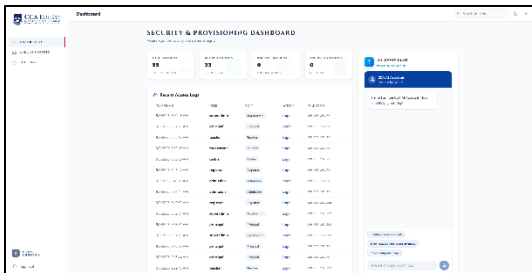


Figure 17.35. Super Admin Dashboard Page.

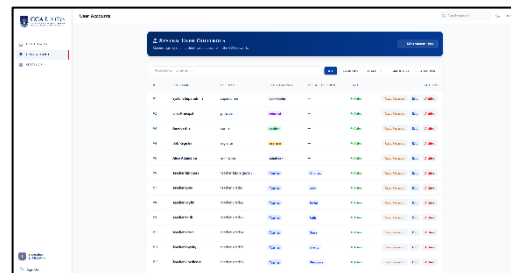


Figure 17.36. Super Admin User Accounts Page.

i. Simulacrum

The simulacrum of the **CCA EduSys: AI-Enhanced School Management System with AI Assistance for Calvary Christian Academy** illustrates how the proposed system addresses the challenges experienced by school personnel in managing student and administrative information. It begins with the school personnel, including the Registrar, Admission, Cashier, Teachers, IT Staff, and Principal, who handle various academic and administrative responsibilities. Currently, CCA uses multiple storage platforms and manual procedures, resulting in fragmented student records, repeated data entry, manual tuition monitoring, and time-consuming report preparation. These challenges increase the workload of school personnel and make it more difficult to maintain, retrieve, and update accurate and organized information efficiently.

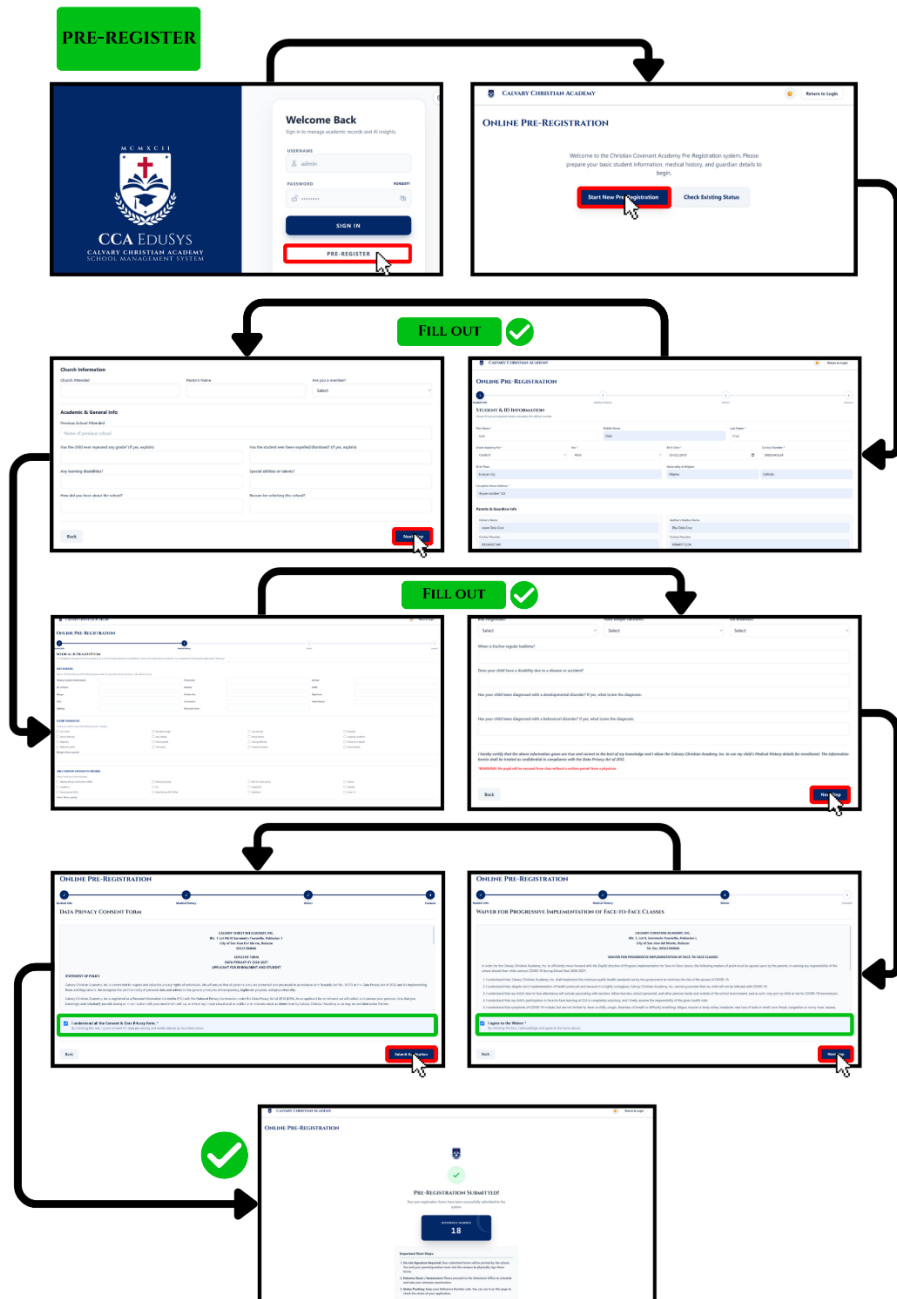


Figure 18.1. Pre-Register Simulacrum.

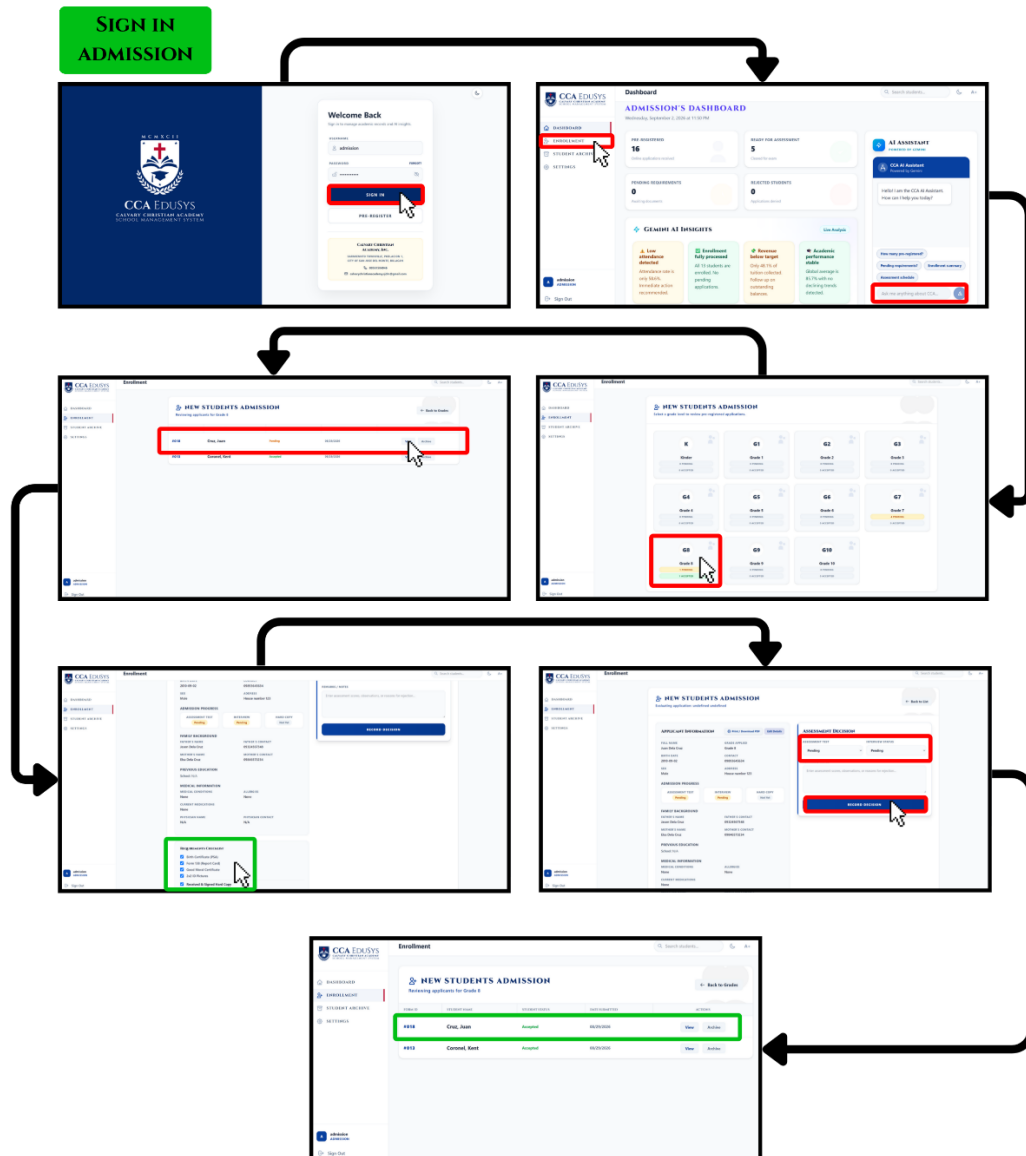


Figure 18.2. Admission Simulacrum.



Figure 18.3. Registrar Simulacrum.

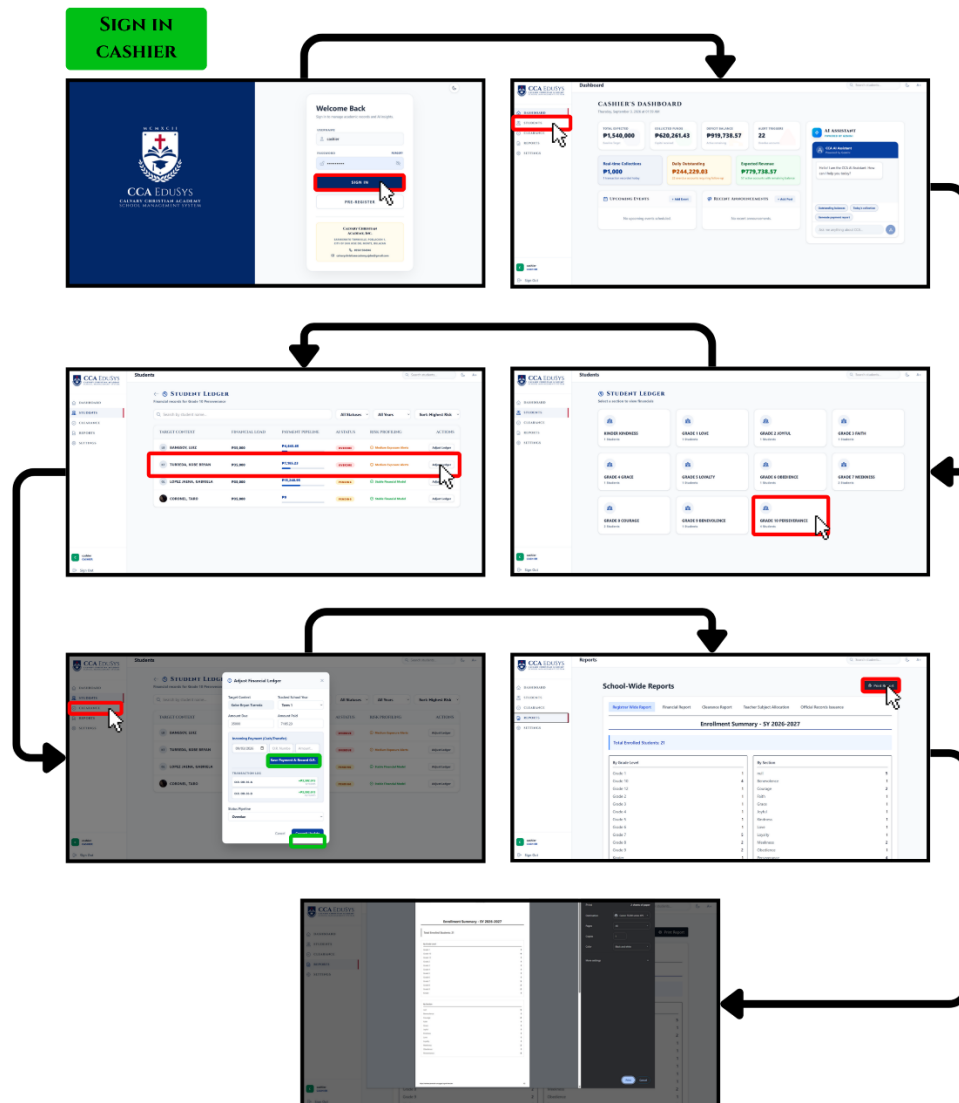


Figure 18.4. Cashier Simulacrum.

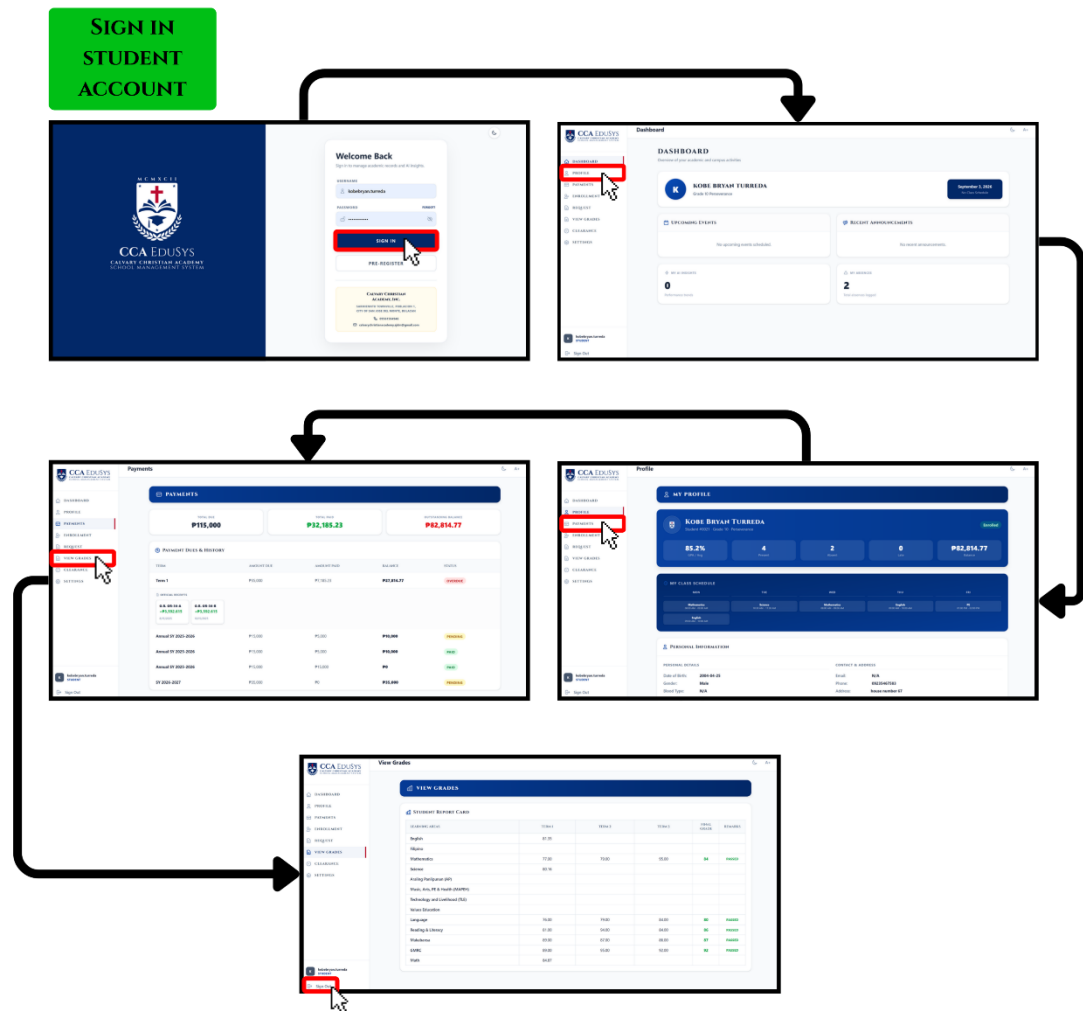


Figure 18.5. Student Simulacrum.

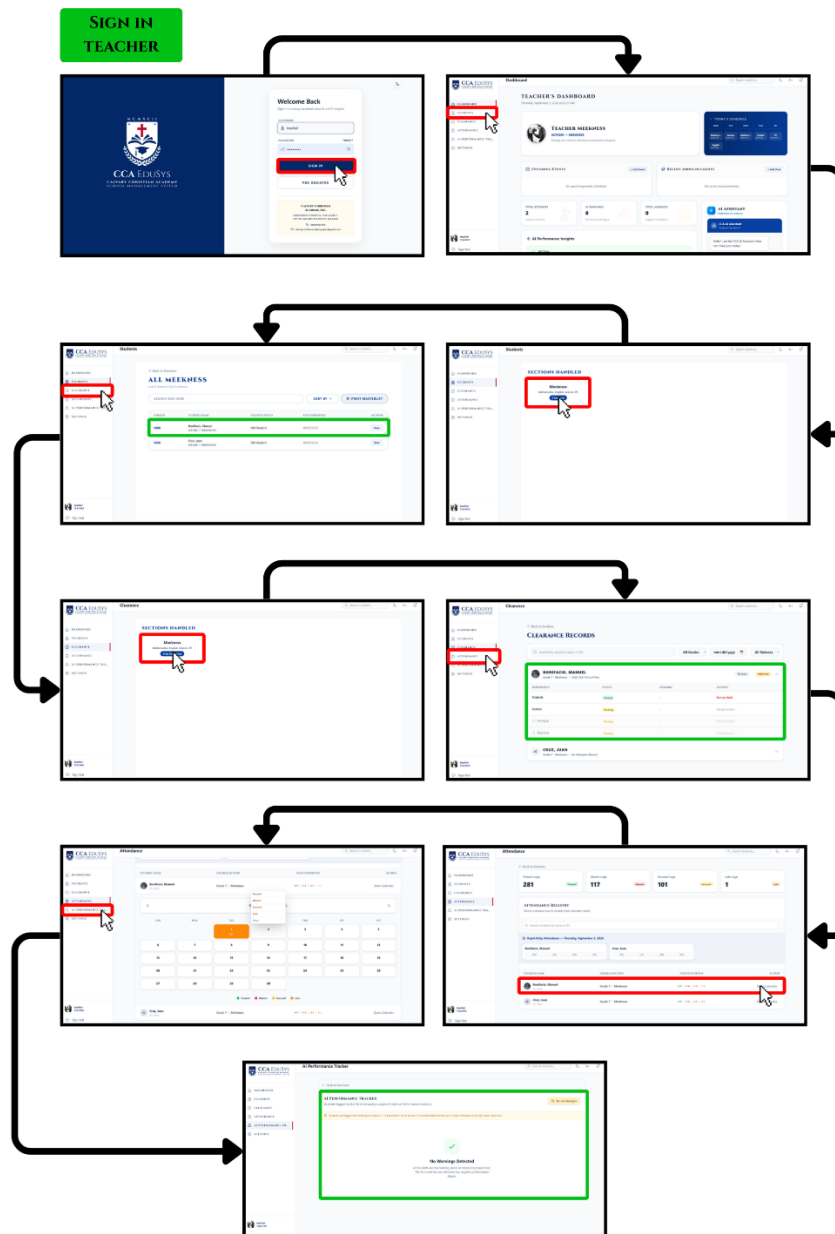


Figure 18.5. Teacher Simulacrum.

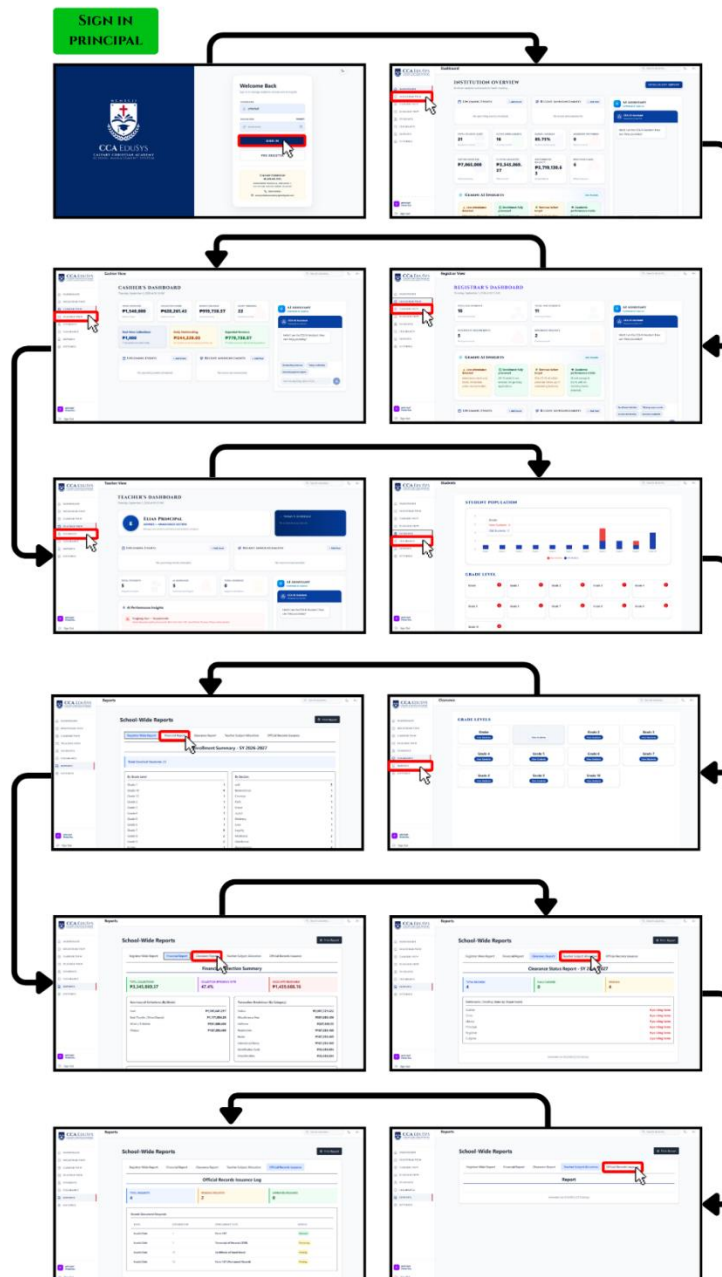


Figure 18.6. *Principal Simulacrum.*



Figure 18.7. Super Admin Simulacrum.

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